

Three-Stage Stochastic Dispatch for Provincial Power System Market Boundary Optimization

Contents

1	Problem Setting	3
1.1	Lead-Time Constraints	3
1.2	Temporal Resolution and Operational Objectives	3
1.3	Uncertainty Structure	4
2	Nomenclature	4
2.1	Sets and Indices	4
2.2	Decision Variables by Stage	4
2.3	Parameters	5
3	Three-Stage Structure and Scenario Tree	7
3.1	Stage Overview	7
3.2	Scenario Tree Construction	7
3.2.1	Stage 1: Root Node (Day D)	7
3.2.2	Stage 2 Branching ($D+1 \sim D+k$)	7
3.2.3	Stage 3 Branching ($D+k+1 \sim D+N$)	8
3.2.4	Joint Scenario Probability	8
3.2.5	Scenario Generation Procedure	8
4	Mathematical Model	9
4.1	Compact Three-Stage Formulation	9
4.2	First-Stage Objective $f(\mathbf{x})$	9
4.3	First-Stage Constraints $\mathcal{X}$	10
4.3.1	ESS Constraints ($d = 1, \dots, k$)	10
4.3.2	Demand-Side Response Constraints ($d = 2, \dots, N$)	10
4.3.3	Hydro Constraints ($d = 3, \dots, N$)	10
4.3.4	Maintenance Schedule Constraints ($d = 1, \dots, N$)	11
4.4	Stage 2: Security-Oriented Dispatch ($\mathbf{y}_{\omega_2}, d = 1, \dots, k$)	11
4.4.1	Stage 2 Constraints	11
4.5	Inter-Stage Coupling (Stage 2 $\rightarrow$ Stage 3)	13
4.5.1	C1: Hydro Reservoir Volume	13
4.5.2	C2: ESS State of Charge	13
4.5.3	C3: Thermal Unit Commitment State	14
4.5.4	Coupling Summary	14
4.6	Stage 3: Adequacy-Oriented Dispatch ($\mathbf{z}_{\omega}, d = k+1, \dots, N$)	14
4.6.1	Stage 3 Constraints	14
4.7	Deterministic Equivalent (Extensive Form)	15

5	Theoretical Properties	16
5.1	Standing Assumptions	16
5.2	LP Relaxation Properties	16
5.3	Value of Stochastic Solution (VSS) Bounds	17
6	Solution Approaches and Convergence Analysis	17
6.1	Direct Solve (Extensive Form)	17
6.2	Nested Benders Decomposition	18
6.2.1	Algorithm Description	18
6.2.2	Convergence of the LP Relaxation	18
6.2.3	The Integer Recourse Challenge	19
6.3	Progressive Hedging	20
6.3.1	Algorithm Description	20
6.3.2	Convergence Properties	20
6.4	Comparison and Practical Recommendations	21
7	Risk-Averse Extension: CVaR	21
8	Summary	22
9	Numerical Verification	22
9.1	Why Week-Scale and Multi-Scenario Are Necessary	22
9.2	Test System	23
9.3	Results	24
9.3.1	Solution Summary	24
9.3.2	Stage-1 Here-and-Now Decisions	25
9.3.3	Line Congestion and Load Shedding	26
9.3.4	Cost and VSS Analysis	27
9.4	Validation of Model Properties	28
9.4.1	Recourse Feasibility	28
9.4.2	Non-Anticipativity	29
9.4.3	Inter-Stage Coupling	29
9.4.4	DC Power Flow Validation	29
9.4.5	Scalability	29
9.5	Decomposition Convergence Analysis	29
9.5.1	Convergence Trajectory	29
9.5.2	Verification of Theoretical Properties	31
9.5.3	Computational Efficiency Comparison	31
9.6	Case Study Design (IEEE Style)	32
9.6.1	Innovation Hypotheses	32
9.6.2	Single-Variable Experimental Matrix	32
9.6.3	Executable Protocol for C0–C4 Reproducibility	33
9.6.4	Weekly Stress-Test Configuration	33
9.7	Managerial Insights	34
9.7.1	Insight 1: Why Multi-Scale Market-Boundary Adjustment Matters	34
9.7.2	Insight 2: Why Variable-Scale Scenario Metrics Are Needed	35
9.7.3	Insight 3: Why Efficient Decoupling Is Operationally Necessary	35
9.7.4	Insight 4: Recommended Planning Workflow	35

1 Problem Setting

The provincial power system under study comprises:

- **Generation:** thermal ($\mathcal{G}$), hydro ($\mathcal{H}$), wind ($\mathcal{W}$), and solar PV ($\mathcal{S}$); all participate in the electricity market with quantity–price bids.
- **Network:** AC transmission grid with monitored lines ($\mathcal{L}$) and sections ($\mathcal{S}_{\text{sec}}$).
- **Load:** controllable large industrial users ($\mathcal{D}^{\text{ctrl}}$) and conventional loads ($\mathcal{D}^{\text{conv}}$).
- **Storage:** pumped-hydro ($\mathcal{E}^{\text{PH}}$) and battery storage ($\mathcal{E}^{\text{BT}}$), collectively $\mathcal{E} = \mathcal{E}^{\text{PH}} \cup \mathcal{E}^{\text{BT}}$; participate as price takers (quantity only, no price bid).

The provincial dispatch centre optimises the market operating boundary for the coming $D+1$ to $D+N$ days (default $N = 7$, i.e. one week ahead) using the following instruments:

1. Adjustment of ESS charge/discharge schedules;
2. Demand-side response (DSR) via invitation of large users;
3. Hydro generation plan adjustment;
4. Maintenance schedule adjustment for thermal units and lines.

1.1 Lead-Time Constraints

Different resources have different minimum lead times for schedule adjustment, as shown in Figure 1:

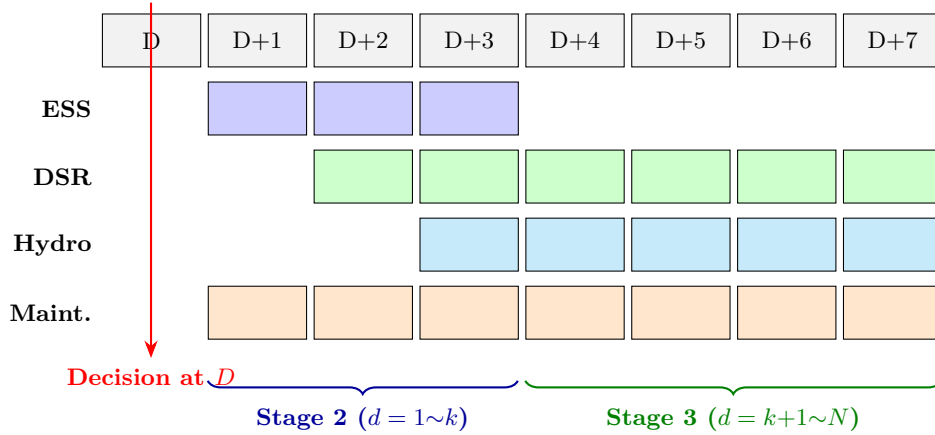


Figure 1: Lead-time structure for adjustable resources ($k = 3$ shown). Coloured cells indicate days whose schedules can be adjusted at day D .

1.2 Temporal Resolution and Operational Objectives

The N -day horizon is partitioned into two operational phases with different modelling fidelity:

Table 1: Three-stage operational structure.

Stage	Days	Objective	Δt	Network model
Stage 1 (Here-and-now)	D	Boundary optimisation	–	–
Stage 2 (Security)	$D+1 \sim D+k$	Operational security	15 min	DC-PF, line limits, nodal I
Stage 3 (Adequacy)	$D+k+1 \sim D+N$	Resource adequacy	1 h	Power/energy balance, zonal

1.3 Uncertainty Structure

- Day $D+1$: renewable generation and conventional load forecasts are *deterministic* (point forecast).
- Days $D+2 \sim D+k$ (remainder of Stage 2): uncertainty $\xi^{(2)}$ is revealed; represented by a finite set of scenarios branching at the beginning of $D+2$.
- Days $D+k+1 \sim D+N$ (Stage 3): additional uncertainty $\xi^{(3)}$ is revealed *conditionally* on the Stage 2 realisation; the scenario tree branches again at the Stage 2/Stage 3 boundary.

2 Nomenclature

2.1 Sets and Indices

Symbol	Description
$\mathcal{D}_N$	Set of future days, $\{1, 2, \dots, N\}$; day index d (day 1 $\equiv D+1$)
$\mathcal{T}_d$	Set of time periods within day d ; $ \mathcal{T}_d = 96$ if $d \leq k$ (15-min), $ \mathcal{T}_d = 24$ if $d > k$ (1-h)
$\mathcal{G}$	Set of thermal generators, indexed by g
$\mathcal{H}$	Set of hydro stations, indexed by h
$\mathcal{W}$	Set of wind farms, indexed by w
$\mathcal{S}$	Set of solar PV plants, indexed by s
$\mathcal{R}$	$\mathcal{W} \cup \mathcal{S}$, all renewables, indexed by r
$\mathcal{E}$	Set of ESS units (pumped-hydro + battery), indexed by e
$\mathcal{D}^{\text{ctrl}}$	Set of controllable (large-user) loads, indexed by j
$\mathcal{D}^{\text{conv}}$	Set of conventional loads
$\mathcal{L}$	Set of monitored AC lines, indexed by l
$\mathcal{B}$	Set of buses (nodes), indexed by b
$\mathcal{M}_g$	Set of bid segments for thermal generator g
<i>Scenario tree sets (see §3.2):</i>	
Ω_2	Set of Stage 2 scenario branches, indexed by ω_2 , $ \Omega_2 = S_2$
$\Omega_3(\omega_2)$	Set of Stage 3 scenario branches <i>conditional on</i> ω_2 , indexed by ω_3 , $ \Omega_3(\omega_2) = S_3(\omega_2)$
Ω	Set of all root-to-leaf scenario paths, $\omega = (\omega_2, \omega_3)$

2.2 Decision Variables by Stage

Stage 1: first-stage decisions $\mathbf{x}$. Made at day D before any uncertainty is revealed.

Variable	Description
<i>ESS schedule (days $1 \leq d \leq k$):</i>	
$P_{e,t,d}^{\text{ch}}$	Charging power of ESS e in period t , day d [MW]
$P_{e,t,d}^{\text{dis}}$	Discharging power of ESS e in period t , day d [MW]
$E_{e,t,d}$	SOC of ESS e at end of period t , day d [MWh]
<i>Demand-side response (days $2 \leq d \leq N$):</i>	

Variable	Description
$P_{j,t,d}^{\text{DR}+}$	Load increase of controllable user j in period t , day d [MW]
$P_{j,t,d}^{\text{DR}-}$	Load reduction of controllable user j in period t , day d [MW]
<i>Hydro plan (days $3 \leq d \leq N$):</i>	
$P_{h,t,d}^H$	Hydro generation of station h in period t , day d [MW]
<i>Maintenance schedule (days $1 \leq d \leq N$):</i>	
$I_{g,d}^G$	Maintenance status of generator g on day d (binary: 1 = under maintenance)
$I_{l,d}^L$	Maintenance status of line l on day d (binary: 1 = under maintenance)

Stage 2: second-stage decisions $\mathbf{y}_{\omega_2}$. Determined after Stage 2 uncertainty $\xi_{\omega_2}^{(2)}$ is revealed, for days $d = 1, \dots, k$.

Variable	Description
P_{g,t,d,ω_2}	Output of thermal g in period t , day d [MW]
$\Delta P_{g,t,m,d,\omega_2}$	Bid segment m dispatch [MW]
u_{g,t,d,ω_2}	Commitment status (binary)
$v_{g,t,d,\omega_2}, w_{g,t,d,\omega_2}$	Startup / shutdown indicators
P_{w,t,d,ω_2}^W	Wind dispatch [MW]
P_{s,t,d,ω_2}^{PV}	Solar dispatch [MW]
$P_{r,t,d,\omega_2}^{\text{cur}}$	Renewable curtailment [MW]
$\text{LS}_{b,t,d,\omega_2}$	Load shedding at bus b (nodal) [MW]
GC_{t,d,ω_2}	Generation curtailment [MW]
F_{l,t,d,ω_2}	Power flow on line l [MW]
θ_{b,t,d,ω_2}	Voltage angle at bus b in period t , day d [rad]

Stage 3: third-stage decisions $\mathbf{z}_{\omega_2,\omega_3}$. Determined after Stage 3 uncertainty $\xi_{\omega_3|\omega_2}^{(3)}$ is revealed, for days $d = k+1, \dots, N$.

Variable	Description
$P_{g,t,d,\omega}$	Output of thermal g [MW], $\omega = (\omega_2, \omega_3)$
$u_{g,t,d,\omega}$	Commitment status (binary)
$P_{w,t,d,\omega}^W, P_{s,t,d,\omega}^{PV}$	Renewable dispatch [MW]
$P_{r,t,d,\omega}^{\text{cur}}$	Renewable curtailment [MW]
$\text{LS}_{t,d,\omega}$	Load shedding (zonal) [MW]
$P_{e,t,d,\omega}^{\text{ch}}, P_{e,t,d,\omega}^{\text{dis}}$	ESS charge/discharge [MW] (free in Stage 3, $d > k$)
$E_{e,t,d,\omega}$	ESS SOC [MWh]

2.3 Parameters

Parameter	Description
N	Planning horizon in days (default 7)
k	Stage 2 / Stage 3 boundary; days $1 \sim k$ are Stage 2
Δt_d	Period length of day d : 0.25 h if $d \leq k$, 1 h if $d > k$
π_{ω_2}	Probability of Stage 2 scenario ω_2
$\pi_{\omega_3 \omega_2}$	Conditional probability of Stage 3 scenario ω_3 given ω_2
$C_{g,m}$	Bid price of thermal g , segment m [¥/MWh]
C_g^U	Startup cost of thermal g [¥]
$C_g^{\min}$	Minimum-output (no-load) cost [¥/h]
C^{LS}	Value of lost load (VOLL) [¥/MWh]
C^{cur}	Renewable curtailment penalty [¥/MWh]
C_j^{DR}	DSR compensation cost for user j [¥/MWh]
$P_{h,d}^{H,\text{ref}}$	Reference (planned) daily hydro energy of station h on day d [MWh]
C_h^{H+}	Cost penalty for hydro energy <i>increase</i> above reference [¥/MWh] (opportunity cost of depleting reservoir)
C_h^{H-}	Cost penalty for hydro energy <i>reduction</i> below reference [¥/MWh]; typically $C_h^{H-} > C_h^{H+}$ reflecting spill risk and water-right constraints
$P_g^{\max}, P_g^{\min}$	Generator capacity / minimum stable output [MW]
$\Delta P_g^U, \Delta P_g^D$	Ramp up/down rate [MW/period]
T_g^U, T_g^D	Minimum up/down time [periods]
$P_e^{\text{ch},\max}, P_e^{\text{dis},\max}$	ESS max charge/discharge power [MW]
$\bar{E}_e, \underline{E}_e$	SOC upper/lower limits [MWh]
$\eta_e^{\text{ch}}, \eta_e^{\text{dis}}$	Charge/discharge efficiency
N_e^{cycle}	Maximum daily cycles
X_l	Reactance of line l [p.u.]
from(l), to(l)	Sending and receiving bus of line l
M^F	Big-M constant for switchable-line DC power flow relaxation (a safe choice is $M^F = 2\pi/X_{\min}$)
b^{ref}	Reference (slack) bus index
$\bar{F}_l$	Thermal capacity of line l [MW]
$\tilde{P}_{r,t,d,\omega}^R$	Renewable available power in scenario ω [MW]
$\tilde{D}_{b,t,d,\omega}$	Load at bus b in scenario ω [MW]
$Q_{h,d}^{\min}, Q_{h,d}^{\max}$	Hydro daily energy limits [MWh]
$\bar{P}_j^{\text{DR}}$	Maximum DR capacity of user j [MW]
$E_{j,d}^{\text{req}}$	Daily energy requirement of controllable user j on day d [MWh]
$T_g^{\text{maint},\min}$	Minimum consecutive maintenance days for unit g

3 Three-Stage Structure and Scenario Tree

3.1 Stage Overview

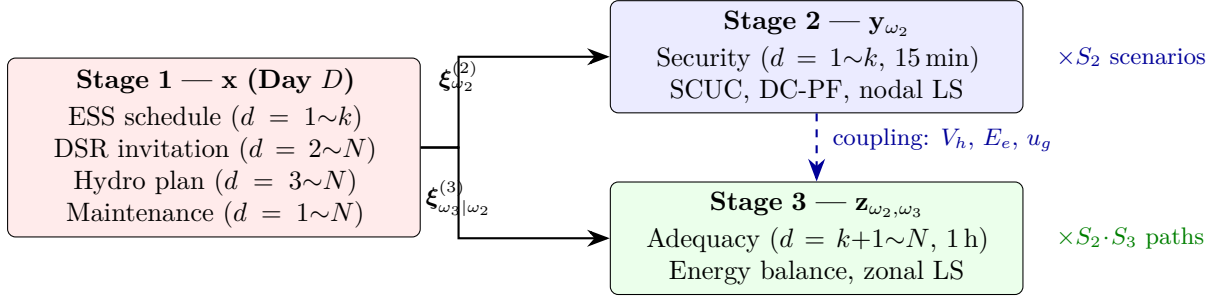


Figure 2: Three-stage structure. Stage 2 and Stage 3 are coupled through reservoir volume, ESS SOC, and thermal unit state at the end of day k .

3.2 Scenario Tree Construction

The uncertainty in the N -day horizon is captured by a discrete *scenario tree* with three levels (Figure 3).

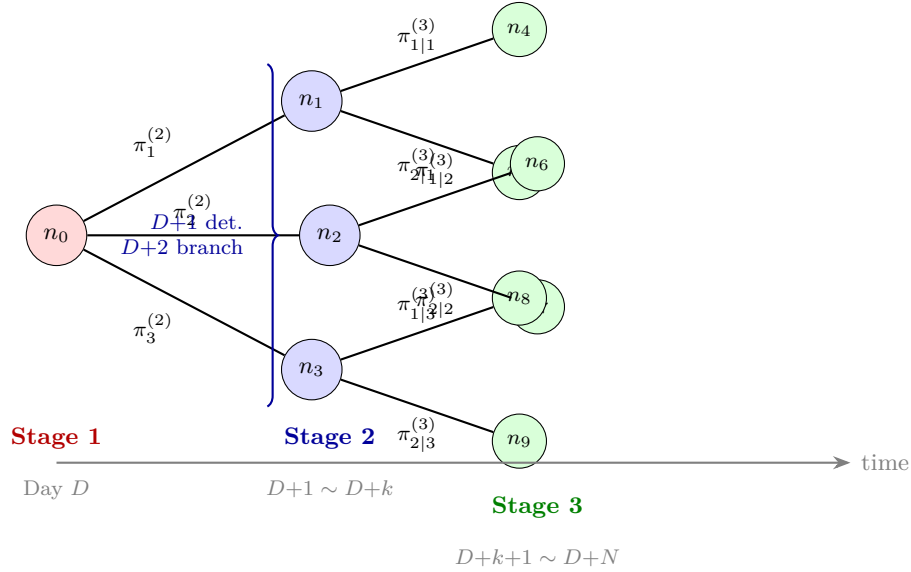


Figure 3: Scenario tree with $S_2 = 3$ Stage 2 branches and $S_3 = 2$ Stage 3 sub-branches per parent, giving 6 leaf (root-to-leaf) paths.

3.2.1 Stage 1: Root Node (Day D)

The root node n_0 represents day D , where all first-stage decisions $\mathbf{x}$ are made under certainty. No branching occurs here.

3.2.2 Stage 2 Branching ($D+1 \sim D+k$)

Day $D+1$ is deterministic: all Stage 2 scenarios share the same renewable and load data for $d = 1$. The tree branches at the beginning of day $D+2$ into S_2 scenarios:

$$\xi_{\omega_2}^{(2)} = \{\tilde{P}_{r,t,d,\omega_2}^R, \tilde{D}_{b,t,d,\omega_2}\}_{r,b,t,d=2,\dots,k} \quad \omega_2 \in \Omega_2 \quad (1)$$

with probabilities $\pi_{\omega_2}^{(2)}$, $\sum_{\omega_2} \pi_{\omega_2}^{(2)} = 1$.

Within each Stage 2 scenario ω_2 , the second-stage decisions $\mathbf{y}_{\omega_2}$ cover *all* days $d = 1, \dots, k$. Since $d = 1$ is shared, we have the *non-anticipativity constraint*:

$$\mathbf{y}_{\omega_2}|_{d=1} = \mathbf{y}_{\omega'_2}|_{d=1} \quad \forall \omega_2, \omega'_2 \in \Omega_2 \quad (2)$$

i.e. day-1 dispatch decisions must be identical across all Stage 2 scenarios because day-1 data is deterministic and decisions for day 1 are made *before* the $D+2$ branching.

3.2.3 Stage 3 Branching ($D+k+1 \sim D+N$)

At the boundary between day k and day $k+1$, each Stage 2 scenario ω_2 further branches into $S_3(\omega_2)$ sub-scenarios:

$$\boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)} = \{\tilde{P}_{r,t,d,\omega}^R, \tilde{D}_{b,t,d,\omega}\}_{r,b,t,d=k+1,\dots,N} \quad \omega_3 \in \Omega_3(\omega_2) \quad (3)$$

with conditional probabilities $\pi_{\omega_3|\omega_2}^{(3)}$, $\sum_{\omega_3} \pi_{\omega_3|\omega_2}^{(3)} = 1$. The third-stage decisions $\mathbf{z}_\omega$ are indexed by the full path $\omega = (\omega_2, \omega_3)$.

3.2.4 Joint Scenario Probability

The probability of leaf scenario $\omega = (\omega_2, \omega_3)$ is:

$$\pi_\omega = \pi_{\omega_2}^{(2)} \cdot \pi_{\omega_3|\omega_2}^{(3)} \quad (4)$$

The total number of leaf scenarios is $|\Omega| = \sum_{\omega_2} S_3(\omega_2)$. If uniform branching is used ($S_3(\omega_2) \equiv S_3$), then $|\Omega| = S_2 \cdot S_3$.

3.2.5 Scenario Generation Procedure

1. **Stage 2 scenarios.** From the day-ahead forecast for $D+2 \sim D+k$, generate S_2 scenarios for renewables and load. Approaches:
 - Sample forecast error vectors from fitted marginal distributions (Beta for solar, truncated Gaussian for wind) with copula-based temporal/spatial correlation.
 - Apply Latin Hypercube Sampling followed by Kantorovich-distance scenario reduction.
2. **Stage 3 scenarios (conditional).** For each Stage 2 scenario ω_2 , generate S_3 sub-scenarios for $D+k+1 \sim D+N$. The conditional generation reflects:
 - *Increased spread*: forecast uncertainty grows with lead time, so Stage 3 scenarios have wider variance than Stage 2.
 - *Conditional dependence*: weather patterns realised in Stage 2 inform the conditional distribution. For example, if ω_2 is a “low-wind” Stage 2 scenario, the Stage 3 sub-scenarios should be drawn from the conditional distribution $P(\boldsymbol{\xi}^{(3)} | \boldsymbol{\xi}^{(2)} = \boldsymbol{\xi}_{\omega_2}^{(2)})$.
3. **Tree balancing (optional).** If uniform branching ($S_3(\omega_2) = S_3$ for all ω_2) is used, the tree is *balanced* and the total number of leaves is $S_2 S_3$. A typical configuration: $S_2 = 5$, $S_3 = 3$, giving 15 leaf scenarios.

Remark on $D+1$ determinism. Although day $D+1$ uses a deterministic forecast, it is part of Stage 2 because the security-oriented SCUC dispatch must be computed jointly for all of $d = 1, \dots, k$. The non-anticipativity constraint (2) ensures that day-1 decisions do not “cheat” by exploiting information from the $D+2$ branching.

4 Mathematical Model

4.1 Compact Three-Stage Formulation

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) + \sum_{\omega_2 \in \Omega_2} \pi_{\omega_2}^{(2)} \left[\mathcal{Q}^{(2)}(\mathbf{x}, \boldsymbol{\xi}_{\omega_2}^{(2)}) + \sum_{\omega_3 \in \Omega_3(\omega_2)} \pi_{\omega_3|\omega_2}^{(3)} \mathcal{Q}^{(3)}(\mathbf{x}, \mathbf{y}_{\omega_2}^*, \boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)}) \right] \quad (5)$$

- $f(\mathbf{x})$: First-stage cost (ESS bids, DSR compensation, hydro adjustment cost; §4.2).
- $\mathcal{Q}^{(2)}(\mathbf{x}, \boldsymbol{\xi}_{\omega_2}^{(2)})$: Optimal dispatch cost for Stage 2 (security) under scenario ω_2 , with first-stage decisions $\mathbf{x}$ fixed.
- $\mathcal{Q}^{(3)}(\mathbf{x}, \mathbf{y}_{\omega_2}^*, \boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)})$: Optimal dispatch cost for Stage 3 (adequacy) under scenario (ω_2, ω_3) , with both $\mathbf{x}$ and the Stage 2 solution $\mathbf{y}_{\omega_2}^*$ fixed. The dependence on $\mathbf{y}_{\omega_2}^*$ arises from *inter-stage coupling* (§4.5).

In the *nested* form, $\mathcal{Q}^{(2)}$ internally optimises over $\mathbf{y}_{\omega_2}$ and passes the terminal state to $\mathcal{Q}^{(3)}$:

$$\mathcal{Q}^{(2)}(\mathbf{x}, \boldsymbol{\xi}_{\omega_2}^{(2)}) = \min_{\mathbf{y}_{\omega_2} \in \mathcal{Y}(\mathbf{x}, \boldsymbol{\xi}_{\omega_2}^{(2)})} \left\{ g(\mathbf{y}_{\omega_2}) + \sum_{\omega_3} \pi_{\omega_3|\omega_2}^{(3)} \mathcal{Q}^{(3)}(\mathbf{x}, \mathbf{y}_{\omega_2}, \boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)}) \right\} \quad (6)$$

where $g(\mathbf{y}_{\omega_2})$ is the Stage 2 operating cost (§4.4).

4.2 First-Stage Objective $f(\mathbf{x})$

$$\begin{aligned} f(\mathbf{x}) = & \underbrace{\sum_{e \in \mathcal{E}} \sum_{d=1}^k \sum_{t \in \mathcal{T}_d} (\lambda_e^{\text{dis}} P_{e,t,d}^{\text{dis}} + \lambda_e^{\text{ch}} P_{e,t,d}^{\text{ch}}) \Delta t_d}_{\text{ESS bid cost}} \\ & + \underbrace{\sum_{j \in \mathcal{D}^{\text{ctrl}}} \sum_{d=2}^N \sum_{t \in \mathcal{T}_d} C_j^{\text{DR}} (P_{j,t,d}^{\text{DR}+} + P_{j,t,d}^{\text{DR}-}) \Delta t_d}_{\text{DSR compensation cost}} \\ & + \underbrace{\sum_{h \in \mathcal{H}} \sum_{d=3}^N [C_h^{H+} \cdot \delta_{h,d}^{H+} + C_h^{H-} \cdot \delta_{h,d}^{H-}]}_{\text{Hydro adjustment cost}} \end{aligned} \quad (7)$$

Hydro adjustment cost. The hydro adjustment cost penalises deviations from the reference (pre-planned) daily energy. Define the daily energy deviation:

$$\sum_{t \in \mathcal{T}_d} P_{h,t,d}^H \Delta t_d - P_{h,d}^{H,\text{ref}} = \delta_{h,d}^{H+} - \delta_{h,d}^{H-} \quad (8)$$

with $\delta_{h,d}^{H+}, \delta_{h,d}^{H-} \geq 0$:

- $\delta_{h,d}^{H+} > 0$: generation *increased* above reference, drawing down the reservoir. Penalised at C_h^{H+} (opportunity cost of depleted water, future generation foregone).
- $\delta_{h,d}^{H-} > 0$: generation *reduced* below reference, either spilling water or deferring generation. Penalised at C_h^{H-} , typically $C_h^{H-} > C_h^{H+}$ because reduction often implies spillage (wasted water) or violation of downstream ecological flow requirements.

4.3 First-Stage Constraints $\mathcal{X}$

4.3.1 ESS Constraints ($d = 1, \dots, k$)

Charge/discharge limits and mutual exclusion:

$$0 \leq P_{e,t,d}^{\text{ch}} \leq P_e^{\text{ch,max}} \quad \forall e, t, d \quad (9)$$

$$0 \leq P_{e,t,d}^{\text{dis}} \leq P_e^{\text{dis,max}} \quad \forall e, t, d \quad (10)$$

$$P_{e,t,d}^{\text{ch}} + P_{e,t,d}^{\text{dis}} \leq \max(P_e^{\text{ch,max}}, P_e^{\text{dis,max}}) \quad \forall e, t, d \quad (11)$$

SOC dynamics:

$$E_{e,t,d} = E_{e,t-1,d} + \eta_e^{\text{ch}} P_{e,t,d}^{\text{ch}} \Delta t_d - \frac{P_{e,t,d}^{\text{dis}}}{\eta_e^{\text{dis}}} \Delta t_d \quad \forall e, t, d \quad (12)$$

$$\underline{E}_e \leq E_{e,t,d} \leq \bar{E}_e \quad \forall e, t, d \quad (13)$$

Inter-day SOC continuity:

$$E_{e,0,d+1} = E_{e,|\mathcal{T}_d|,d} \quad \forall e, d = 1, \dots, k-1 \quad (14)$$

$$E_{e,0,1} = E_e^{\text{ini}} \quad (15)$$

Cycle limit (per day):

$$\sum_{t \in \mathcal{T}_d} \left(\frac{P_{e,t,d}^{\text{dis}}}{\eta_e^{\text{dis}}} + \eta_e^{\text{ch}} P_{e,t,d}^{\text{ch}} \right) \Delta t_d \leq 2N_e^{\text{cycle}} \bar{E}_e \quad \forall e, d \quad (16)$$

4.3.2 Demand-Side Response Constraints ($d = 2, \dots, N$)

$$0 \leq P_{j,t,d}^{\text{DR}+} \leq \bar{P}_j^{\text{DR}}, \quad 0 \leq P_{j,t,d}^{\text{DR}-} \leq \bar{P}_j^{\text{DR}} \quad \forall j, t, d \quad (17)$$

$$\sum_{t \in \mathcal{T}_d} (D_{j,t,d}^0 + P_{j,t,d}^{\text{DR}+} - P_{j,t,d}^{\text{DR}-}) \Delta t_d = E_{j,d}^{\text{req}} \quad \forall j, d \quad (18)$$

4.3.3 Hydro Constraints ($d = 3, \dots, N$)

Output limits:

$$P_h^{H,\min} \leq P_{h,t,d}^H \leq P_h^{H,\max} \quad \forall h, t, d \quad (19)$$

Daily energy limits:

$$Q_{h,d}^{\min} \leq \sum_{t \in \mathcal{T}_d} P_{h,t,d}^H \Delta t_d \leq Q_{h,d}^{\max} \quad \forall h, d \quad (20)$$

Multi-day reservoir water balance:

$$V_{h,d} = V_{h,d-1} + I_{h,d} - \sum_{t \in \mathcal{T}_d} \frac{P_{h,t,d}^H}{\rho_h} \Delta t_d - S_{h,d} \quad \forall h, d \quad (21)$$

$$\underline{V}_h \leq V_{h,d} \leq \bar{V}_h \quad \forall h, d \quad (22)$$

Deviation decomposition (8):

$$\delta_{h,d}^{H+}, \delta_{h,d}^{H-} \geq 0 \quad \forall h, d \quad (23)$$

4.3.4 Maintenance Schedule Constraints ($d = 1, \dots, N$)

$$I_{g,d}^G \in \{0, 1\} \quad \forall g, d \quad (24)$$

$$I_{l,d}^L \in \{0, 1\} \quad \forall l, d \quad (25)$$

Minimum consecutive maintenance duration:

$$\sum_{\tau=d}^{\min(d+T_g^{\text{maint},\min}-1, N)} I_{g,\tau}^G \geq T_g^{\text{maint},\min} \cdot (I_{g,d}^G - I_{g,d-1}^G) \quad \forall g, d \geq 2 \quad (26)$$

Irrevocable maintenance:

$$I_{g,d}^G = 1 \quad \forall (g, d) \in \mathcal{F}^{\text{maint}}, \quad I_{l,d}^L = 1 \quad \forall (l, d) \in \mathcal{F}_L^{\text{maint}} \quad (27)$$

4.4 Stage 2: Security-Oriented Dispatch ($\mathbf{y}_{\omega_2}$, $d = 1, \dots, k$)

For a given $\mathbf{x}^*$ and scenario ω_2 :

$$g(\mathbf{y}_{\omega_2}) = \sum_{d=1}^k \sum_{t \in \mathcal{T}_d} \left[\sum_{g \in \mathcal{G}} (C_g(P_{g,t,d,\omega_2}) + C_g^U v_{g,t,d,\omega_2} + C_g^{\min} u_{g,t,d,\omega_2} \Delta t_d) + C^{\text{LS}} \sum_{b \in \mathcal{B}} \text{LS}_{b,t,d,\omega_2} \Delta t_d + C^{\text{cur}} \sum_{r \in \mathcal{R}} P_{r,t,d,\omega_2}^{\text{cur}} \Delta t_d \right] \quad (28)$$

4.4.1 Stage 2 Constraints

S2-1: Nodal power balance. For each bus b , period t , day $d \leq k$:

$$\begin{aligned} \sum_{g \in \mathcal{G}_b} P_{g,t,d,\omega_2} + \sum_{h \in \mathcal{H}_b} \underbrace{P_{h,t,d}^H}_{\text{from } \mathbf{x}} + \sum_{r \in \mathcal{R}_b} (\tilde{P}_{r,t,d,\omega_2}^R - P_{r,t,d,\omega_2}^{\text{cur}}) + \sum_{e \in \mathcal{E}_b} \underbrace{(P_{e,t,d}^{\text{dis}} - P_{e,t,d}^{\text{ch}})}_{\text{from } \mathbf{x}} \\ + \text{LS}_{b,t,d,\omega_2} - \text{GC}_{t,d,\omega_2} = \tilde{D}_{b,t,d,\omega_2} - \underbrace{(P_{j,t,d}^{\text{DR}-} - P_{j,t,d}^{\text{DR}+})}_{\text{from } \mathbf{x}} + \sum_{l:b=\text{to}(l)} F_{l,t,d,\omega_2} - \sum_{l:b=\text{from}(l)} F_{l,t,d,\omega_2} \end{aligned} \quad (29)$$

S2-2: DC power flow with switchable lines (angle-based, big-M). When the network topology changes due to line maintenance, the classical PTDF matrix $G_{l,b}$ must be recomputed for every topology state, coupling the binary maintenance variable $I_{l,d}^L$ to the network model in a non-trivial way. We instead adopt the *angle-based* DC power flow with a big-M relaxation, which naturally embeds the on/off switching of each line into the formulation without requiring topology-dependent PTDF matrices.

Define the DC power flow relationship on line $l = (\text{from}(l), \text{to}(l))$ with reactance X_l :

$$F_l^{\text{dc}} \triangleq \frac{\theta_{\text{from}(l),t,d,\omega_2} - \theta_{\text{to}(l),t,d,\omega_2}}{X_l} \quad (30)$$

When line l is in service ($I_{l,d}^L = 0$), the flow must equal the DC power flow; when line l is under maintenance ($I_{l,d}^L = 1$), the constraint is relaxed by big-M:

$$\boxed{-(1 - I_{l,d}^L) M^F \leq F_{l,t,d,\omega_2} - \frac{\theta_{\text{from}(l),t,d,\omega_2} - \theta_{\text{to}(l),t,d,\omega_2}}{X_l} \leq (1 - I_{l,d}^L) M^F \quad \forall l, t, d \leq k} \quad (31)$$

When $I_{l,d}^L = 0$, the big-M terms vanish and (31) reduces to the exact DC power flow $F_l = (\theta_i - \theta_j)/X_l$. When $I_{l,d}^L = 1$, the right-hand side becomes $\pm M^F$, effectively decoupling F_l from the angle difference; together with the line capacity constraint (33) below (which forces $F_l = 0$ when the line is out), this correctly models the open-circuit condition.

S2-2a: Reference bus angle.

$$\theta_{b^{\text{ref}},t,d,\omega_2} = 0 \quad \forall t, d \leq k, \omega_2 \quad (32)$$

S2-3: Line capacity (with maintenance coupling).

$$-\bar{F}_l(1 - I_{l,d}^L) \leq F_{l,t,d,\omega_2} \leq \bar{F}_l(1 - I_{l,d}^L) \quad \forall l, t, d \leq k \quad (33)$$

When $I_{l,d}^L = 1$, this forces $F_{l,t,d,\omega_2} = 0$, i.e. no power flows through a line under maintenance. Combined with (31), the joint effect is: (i) the angle-flow relationship is deactivated, and (ii) the physical flow is zeroed out.

Remark (Choice of M^F). The big-M constant must be large enough to never be active when $I_{l,d}^L = 0$. A safe and tight choice is $M^F = \max_l \bar{F}_l + \epsilon$ or, equivalently, $M^F = 2\pi/X_{\min}$, where $X_{\min} = \min_l X_l$. An overly large M^F weakens the LP relaxation, so the tightest valid bound should be used.

S2-4: Generator output limits (with maintenance).

$$P_g^{\min} \cdot u_{g,t,d,\omega_2} \leq P_{g,t,d,\omega_2} \leq P_g^{\max} \cdot u_{g,t,d,\omega_2} \cdot (1 - I_{g,d}^G) \quad \forall g, t, d \quad (34)$$

S2-5: Ramp rates.

$$P_{g,t,d,\omega_2} - P_{g,t-1,d,\omega_2} \leq \Delta P_g^U + P_g^{\max} v_{g,t,d,\omega_2} \quad (35)$$

$$P_{g,t-1,d,\omega_2} - P_{g,t,d,\omega_2} \leq \Delta P_g^D + P_g^{\max} w_{g,t,d,\omega_2} \quad (36)$$

S2-6: Minimum up/down time and commitment transition.

$$u_{g,t,d,\omega_2} - u_{g,t-1,d,\omega_2} = v_{g,t,d,\omega_2} - w_{g,t,d,\omega_2} \quad (37)$$

$$\sum_{\tau=\max(1,t-T_g^U+1)}^t v_{g,\tau,d,\omega_2} \leq u_{g,t,d,\omega_2} \quad (38)$$

$$\sum_{\tau=\max(1,t-T_g^D+1)}^t w_{g,\tau,d,\omega_2} \leq 1 - u_{g,t,d,\omega_2} \quad (39)$$

S2-7: Renewable bounds.

$$0 \leq P_{r,t,d,\omega_2}^{\text{cur}} \leq \tilde{P}_{r,t,d,\omega_2}^R \quad \forall r, t, d \quad (40)$$

S2-8: Reserve requirements.

$$\sum_g P_g^{\max} (1 - I_{g,d}^G) u_{g,t,d,\omega_2} - \sum_g P_{g,t,d,\omega_2} \geq R_{t,d}^{\text{up}} \quad (41)$$

$$\sum_g P_{g,t,d,\omega_2} - \sum_g P_g^{\min} u_{g,t,d,\omega_2} \geq R_{t,d}^{\text{dn}} \quad (42)$$

S2-9: Non-anticipativity for day 1.

$$\mathbf{y}_{\omega_2}|_{d=1} = \mathbf{y}_{\omega'_2}|_{d=1} \quad \forall \omega_2, \omega'_2 \in \Omega_2 \quad (43)$$

4.5 Inter-Stage Coupling (Stage 2 $\rightarrow$ Stage 3)

The terminal state of Stage 2 under scenario ω_2 defines the *initial conditions* for Stage 3. This coupling is the key structural feature distinguishing the three-stage model from two independent two-stage problems.

4.5.1 C1: Hydro Reservoir Volume

The reservoir volume at the end of day k (last day of Stage 2) feeds into Stage 3 as the initial volume for day $k+1$:

$$V_{h,k,\omega_2}^{(3)} = V_{h,k-1} + I_{h,k} - \sum_{t \in \mathcal{T}_k} \frac{P_{h,t,k}^H}{\rho_h} \Delta t_k - S_{h,k} \quad (44)$$

Here $V_{h,k-1}$ and $P_{h,t,k}^H$ come from the first-stage hydro plan, while the resulting $V_{h,k,\omega_2}^{(3)}$ becomes a *parameter* in Stage 3 constraints. Note that although $P_{h,t,d}^H$ for $d \leq k$ is a first-stage decision (and thus deterministic), the reservoir may also be affected by stochastic inflow; in that case $V_{h,k,\omega_2}^{(3)}$ depends on ω_2 through I_{h,d,ω_2} .

Within Stage 3, the hydro reservoir evolves as:

$$V_{h,d,\omega} = V_{h,d-1,\omega} + I_{h,d,\omega} - \sum_{t \in \mathcal{T}_d} \frac{P_{h,t,d}^H}{\rho_h} \Delta t_d - S_{h,d,\omega} \quad d = k+1, \dots, N \quad (45)$$

with $V_{h,k,\omega} = V_{h,k,\omega_2}^{(3)}$ as the initial condition.

4.5.2 C2: ESS State of Charge

The SOC of each ESS at the end of Stage 2 (end of day k) becomes the initial SOC for Stage 3:

$$E_{e,0,\omega_2}^{(3)} = E_{e,|\mathcal{T}_k|,k} \quad (46)$$

Since the ESS schedule for $d \leq k$ is a first-stage decision, $E_{e,0,\omega_2}^{(3)}$ is the same across all ω_2 . However, it links first-stage ESS decisions to Stage 3 feasibility: an aggressive discharge in Stage 2 leaves less stored energy for Stage 3.

In Stage 3 ($d > k$), ESS dispatch becomes a *recourse decision*, with SOC dynamics:

$$E_{e,t,d,\omega} = E_{e,t-1,d,\omega} + \eta_e^{\text{ch}} P_{e,t,d,\omega}^{\text{ch}} \Delta t_d - \frac{P_{e,t,d,\omega}^{\text{dis}}}{\eta_e^{\text{dis}}} \Delta t_d \quad (47)$$

with inter-day continuity $E_{e,0,d+1,\omega} = E_{e,|\mathcal{T}_d|,d,\omega}$ for $d = k+1, \dots, N-1$, and:

$$E_{e,0,k+1,\omega} = E_{e,0,\omega_2}^{(3)} \quad (48)$$

4.5.3 C3: Thermal Unit Commitment State

The commitment status of each thermal unit at the *last period* of day k determines the initial state for Stage 3:

$$u_{g,0,\omega_2}^{(3)} = u_{g,|\mathcal{T}_k|,k,\omega_2} \quad (49)$$

This coupling is scenario-dependent: different Stage 2 scenarios may lead to different commitment schedules at the end of day k , and Stage 3 must respect the minimum up/down time constraints spanning the Stage 2/3 boundary:

$$u_{g,1,k+1,\omega} - u_{g,0,\omega_2}^{(3)} = v_{g,1,k+1,\omega} - w_{g,1,k+1,\omega} \quad (50)$$

Additionally, the number of consecutive on/off periods at the end of Stage 2 must be tracked. Let $\tau_{g,\omega_2}^{\text{on}}$ and $\tau_{g,\omega_2}^{\text{off}}$ be the number of consecutive on/off periods at the end of day k in scenario ω_2 . The minimum up/down time constraints at the start of Stage 3 become:

$$u_{g,t,k+1,\omega} = 1 \quad \forall t \leq T_g^U - \tau_{g,\omega_2}^{\text{on}} \text{ if } u_{g,0,\omega_2}^{(3)} = 1 \text{ and } \tau_{g,\omega_2}^{\text{on}} < T_g^U \quad (51)$$

$$u_{g,t,k+1,\omega} = 0 \quad \forall t \leq T_g^D - \tau_{g,\omega_2}^{\text{off}} \text{ if } u_{g,0,\omega_2}^{(3)} = 0 \text{ and } \tau_{g,\omega_2}^{\text{off}} < T_g^D \quad (52)$$

4.5.4 Coupling Summary

Table 7: Inter-stage coupling: Stage 2 terminal state $\rightarrow$ Stage 3 initial condition.

State variable	End of Stage 2	Start of Stage 3	Depends on ω_2 ?
Reservoir volume	$V_{h,k}$ (from $\mathbf{x}$)	$V_{h,k,\omega_2}^{(3)}$ (44)	Only if inflow is stochastic
ESS SOC	$E_{e, \mathcal{T}_k ,k}$ (from $\mathbf{x}$)	$E_{e,0,\omega_2}^{(3)}$ (46)	No (deterministic from S1)
Thermal status	$u_{g, \mathcal{T}_k ,k,\omega_2}$ (from $\mathbf{y}$)	$u_{g,0,\omega_2}^{(3)}$ (49)	Yes
Thermal on/off time	$\tau_{g,\omega_2}^{\text{on/off}}$	Min up/down init (51)–(52)	Yes

4.6 Stage 3: Adequacy-Oriented Dispatch ($\mathbf{z}_\omega$, $d = k+1, \dots, N$)

For given $\mathbf{x}^*$, $\mathbf{y}_{\omega_2}^*$, and scenario $\omega = (\omega_2, \omega_3)$:

$$\mathcal{Q}^{(3)}(\mathbf{x}, \mathbf{y}_{\omega_2}, \boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)}) = \min_{\mathbf{z}_\omega} \sum_{d=k+1}^N \sum_{t \in \mathcal{T}_d} \left[\sum_{g \in \mathcal{G}} (C_g(P_{g,t,d,\omega}) + C_g^{\min} u_{g,t,d,\omega} \Delta t_d) + C^{\text{LS}} \text{LS}_{t,d,\omega} \Delta t_d + C^{\text{cur}} \sum_r P_{r,t,d,\omega}^{\text{cur}} \Delta t_d \right] \quad (53)$$

4.6.1 Stage 3 Constraints

S3-1: System-wide power balance (no DC-PF).

$$\begin{aligned} & \sum_{g \in \mathcal{G}} P_{g,t,d,\omega} + \sum_{h \in \mathcal{H}} P_{h,t,d}^H + \sum_{r \in \mathcal{R}} (\tilde{P}_{r,t,d,\omega}^R - P_{r,t,d,\omega}^{\text{cur}}) + \sum_{e \in \mathcal{E}} (P_{e,t,d,\omega}^{\text{dis}} - P_{e,t,d,\omega}^{\text{ch}}) \\ & + \text{LS}_{t,d,\omega} = \sum_b \tilde{D}_{b,t,d,\omega} - \sum_j (P_{j,t,d}^{\text{DR}-} - P_{j,t,d}^{\text{DR}+}) \end{aligned} \quad (54)$$

Note: hydro (P^H) and DSR are from $\mathbf{x}$; ESS in Stage 3 is a recourse decision.

S3-2: Generator limits (with maintenance).

$$P_g^{\min} u_{g,t,d,\omega} \leq P_{g,t,d,\omega} \leq P_g^{\max} u_{g,t,d,\omega} (1 - I_{g,d}^G) \quad \forall g, t, d > k \quad (55)$$

S3-3: Commitment transition with Stage 2 initial state.

$$u_{g,t,d,\omega} - u_{g,t-1,d,\omega} = v_{g,t,d,\omega} - w_{g,t,d,\omega} \quad t \geq 2 \quad (56)$$

For $t = 1$ of day $k+1$, use (50). The minimum up/down time constraints (51)–(52) apply at the Stage 2/3 boundary.

S3-4: ESS constraints in Stage 3. ESS dispatch is a Stage 3 recourse decision with:

$$0 \leq P_{e,t,d,\omega}^{\text{ch}} \leq P_e^{\text{ch,max}} \quad (57)$$

$$0 \leq P_{e,t,d,\omega}^{\text{dis}} \leq P_e^{\text{dis,max}} \quad (58)$$

SOC dynamics (47), bounds $\underline{E}_e \leq E_{e,t,d,\omega} \leq \bar{E}_e$, and initial condition (48) from Stage 2 coupling.

S3-5: Hydro reservoir coupling with Stage 2. Water balance (45) with initial condition $V_{h,k,\omega} = V_{h,k,\omega_2}^{(3)}$ from (44), and volume bounds $\underline{V}_h \leq V_{h,d,\omega} \leq \bar{V}_h$.

S3-6: Daily power and energy adequacy.

$$\sum_g P_g^{\max} (1 - I_{g,d}^G) + \sum_h P_h^{H,\max} + \sum_e P_e^{\text{dis,max}} \geq \max_t \sum_b \tilde{D}_{b,t,d,\omega} + R_d^{\text{margin}} \quad (59)$$

4.7 Deterministic Equivalent (Extensive Form)

The full three-stage deterministic equivalent is:

$$\boxed{\min_{\mathbf{x}, \{\mathbf{y}_{\omega_2}\}, \{\mathbf{z}_{\omega}\}} f(\mathbf{x}) + \sum_{\omega_2} \pi_{\omega_2}^{(2)} \left[g(\mathbf{y}_{\omega_2}) + \sum_{\omega_3} \pi_{\omega_3|\omega_2}^{(3)} \mathcal{Q}^{(3)}(\mathbf{x}, \mathbf{y}_{\omega_2}, \boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)}) \right]} \quad (60)$$

subject to:

First-stage ($\mathbf{x}$) : (9)–(27), (8)–(23)

Stage 2 ($\mathbf{y}_{\omega_2}$) $\forall \omega_2$: (29)–(43)

Stage 2→3 coupling $\forall \omega_2$: (44)–(52)

Stage 3 ($\mathbf{z}_{\omega}$) $\forall \omega = (\omega_2, \omega_3)$: (54)–(59)

Problem size. With S_2 Stage 2 scenarios and S_3 Stage 3 sub-branches per parent, the extensive form has:

- n_x first-stage variables;
- $S_2 \cdot n_y$ Stage 2 variables (but day-1 shared via non-anticipativity, effectively $n_y^{(d=1)} + S_2 \cdot n_y^{(d=2..k)}$);
- $S_2 \cdot S_3 \cdot n_z$ Stage 3 variables.

Typical configuration: $k = 3$, $N = 7$, $S_2 = 5$, $S_3 = 3$ gives 15 leaf scenarios. Stage 2 dominates in per-scenario complexity (96 periods/day $\times$ 3 days with SCUC binaries).

5 Theoretical Properties

This section establishes the formal assumptions under which the three-stage formulation (5) is well-posed and analyses the structural properties that underpin both the value of stochastic programming and the convergence of decomposition algorithms.

5.1 Standing Assumptions

Assumption 1 (Finite scenario tree). *The scenario tree has finite branching: $|\Omega_2| = S_2 < \infty$ and $|\Omega_3(\omega_2)| = S_3(\omega_2) < \infty$ for every $\omega_2 \in \Omega_2$, so the total number of leaf scenarios $|\Omega| = \sum_{\omega_2} S_3(\omega_2) < \infty$.*

Assumption 2 (Bounded feasible sets). *The feasible sets $\mathcal{X}$, $\mathcal{Y}(\mathbf{x}, \boldsymbol{\xi}_{\omega_2}^{(2)})$, and $\mathcal{Z}(\mathbf{x}, \mathbf{y}_{\omega_2}, \boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)})$ are non-empty, closed, and bounded for every feasible first-stage decision and every scenario realisation. This holds naturally, since all generation, storage, and flow variables are bounded by installed capacities.*

Assumption 3 (Relatively complete recourse). *For every $\mathbf{x} \in \mathcal{X}$ and every scenario path $\omega = (\omega_2, \omega_3) \in \Omega$, the recourse functions $Q^{(2)}(\mathbf{x}, \boldsymbol{\xi}_{\omega_2}^{(2)})$ and $Q^{(3)}(\mathbf{x}, \mathbf{y}_{\omega_2}, \boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)})$ are finite-valued.*

Proposition 1 (Relatively complete recourse holds). *Assumption 3 is satisfied by construction in our model.*

Proof. Each stage includes *safety-valve* variables with the following hierarchy of penalty costs: $C^{\text{LS}} \gg C^{\text{cur}} \gg \text{fuel cost}$:

1. **Load shedding (LS)**: absorbs any demand shortfall at VOLL penalty C^{LS} .
2. **Renewable curtailment** (P^{cur}): absorbs any excess renewable supply at penalty C^{cur} .
3. **Generation curtailment** (GC): absorbs any over-generation from committed thermal units.

Given Assumption 2, for any $\mathbf{x} \in \mathcal{X}$ and any scenario ω , the Stage 2 power balance (29) can always be satisfied: if total generation falls short of demand, LS absorbs the deficit; if generation exceeds demand, P^{cur} or GC absorbs the surplus. Hence $\mathcal{Y}(\mathbf{x}, \boldsymbol{\xi}_{\omega_2}^{(2)}) \neq \emptyset$. The same argument applies to Stage 3 via (54), establishing finiteness of $Q^{(3)}$. Since $Q^{(2)}$ is the minimum of a finite sum including $Q^{(3)}$ terms, it too is finite. $\square$

5.2 LP Relaxation Properties

Let $\bar{Q}^{(t)}$ denote the *LP relaxation* of the Stage- t value function, obtained by relaxing all binary variables $(u_g, v_g, w_g, I_g^G, I_l^L)$ to $[0, 1]$.

Proposition 2 (Convexity and Lipschitz continuity of LP value function). *Under Assumptions 1–3, the LP relaxation value functions $\bar{Q}^{(2)}(\mathbf{x}, \boldsymbol{\xi}_{\omega_2}^{(2)})$ and $\bar{Q}^{(3)}(\mathbf{x}, \mathbf{y}_{\omega_2}, \boldsymbol{\xi}_{\omega_3|\omega_2}^{(3)})$ are:*

1. *convex and piecewise linear in the coupling variables $(\mathbf{x})$ and $(\mathbf{x}, \mathbf{y}_{\omega_2})$ respectively;*
2. *Lipschitz continuous on $\mathcal{X} \times \mathcal{Y}$ with constant $L \leq C^{\text{LS}}/\Delta t_{\min}$.*

Proof. (i) Both $\bar{Q}^{(2)}$ and $\bar{Q}^{(3)}$ are defined as the optimal value of a parametric LP in which the coupling variables enter only the right-hand side of the linking constraints (44)–(49). By the classical result on parametric LP (see Birge and Louveaux [1997], Theorem 3.1), the optimal value is a convex, piecewise linear function of the right-hand side, provided the feasible set is non-empty (guaranteed by Proposition 1).

(ii) The Lipschitz constant is bounded by the maximum absolute value of the dual variables on the coupling constraints. Since LS provides a universal recourse action at cost C^{LS} per MWh, any unit change in the coupling state (e.g., 1 MWh change in terminal SOC E_e) can be

compensated by at most $1/\Delta t_{\min}$ MW of load-shedding adjustment, costing at most $C^{\text{LS}}/\Delta t_{\min}$. Hence $\|\lambda^*\|_{\infty} \leq C^{\text{LS}}/\Delta t_{\min}$, which bounds the Lipschitz constant via the LP sensitivity theorem. $\square$

Corollary 1 (Expected recourse convexity). *The expected LP recourse function*

$$\bar{Q}(\mathbf{x}) = \sum_{\omega_2} \pi_{\omega_2}^{(2)} \bar{Q}^{(2)}(\mathbf{x}, \xi_{\omega_2}^{(2)})$$

is convex and piecewise linear in $\mathbf{x}$, being a non-negative weighted sum of convex functions.

5.3 Value of Stochastic Solution (VSS) Bounds

We formalise the classical VSS inequalities adapted to the three-stage setting. Define:

- **EV** (Expected Value problem): solve (5) with all scenarios replaced by their expectation $\bar{\xi} = \mathbb{E}[\xi]$; optimal value z^{EV} , optimal first-stage $\mathbf{x}^{\text{EV}}$.
- **EEV** (Expected result of using EV): evaluate $\mathbf{x}^{\text{EV}}$ under the full stochastic model (5); value z^{EEV} .
- **RP** (Recourse Problem): the true stochastic optimum of (5); value z^{RP} .
- **WS** (Wait-and-See): solve a separate deterministic problem for each ω and take the expectation; value z^{WS} .
- **EVPI** (Expected Value of Perfect Information): $z^{\text{EVPI}} = z^{\text{RP}} - z^{\text{WS}}$.
- **VSS**: $z^{\text{VSS}} = z^{\text{EEV}} - z^{\text{RP}}$.

Theorem 1 (Fundamental stochastic programming bounds). *Under Assumptions 1–3:*

$$z^{\text{EV}} \leq z^{\text{WS}} \leq z^{\text{RP}} \leq z^{\text{EEV}} \quad (61)$$

Consequently, $\text{VSS} = z^{\text{EEV}} - z^{\text{RP}} \geq 0$ and $\text{EVPI} = z^{\text{RP}} - z^{\text{WS}} \geq 0$.

Proof. The chain (61) is the Madansky–Edmundson–Jensen inequality adapted to multi-stage programs (Birge and Louveaux [1997], Theorem 4.3). We verify each inequality:

$z^{\text{EV}} \leq z^{\text{WS}}$: The EV problem uses $\bar{\xi}$, which is a feasible “scenario” for a single wait-and-see problem. By Jensen’s inequality applied to the convex recourse function (Corollary 1 for the LP relaxation; the inequality extends to the MILP case because $f + \mathbb{E}[Q] \leq f + Q(\mathbb{E}[\xi])$ holds whenever Q is convex in ξ), the optimal value under the mean scenario underestimates the average of the scenario-specific optima.

$z^{\text{WS}} \leq z^{\text{RP}}$: Wait-and-see uses scenario-specific first-stage decisions, which is a relaxation of the RP where $\mathbf{x}$ must be the same across all scenarios (non-anticipativity). Relaxing a constraint can only improve (or maintain) the objective.

$z^{\text{RP}} \leq z^{\text{EEV}}$: The EEV fixes the first-stage decision to $\mathbf{x}^{\text{EV}}$, while the RP optimises over all $\mathbf{x} \in \mathcal{X}$. Since $\mathbf{x}^{\text{EV}} \in \mathcal{X}$, the RP has a larger feasible set and thus a smaller (or equal) optimal value. $\square$

Remark 1 (Interpretation of VSS). $\text{VSS}/z^{\text{EEV}}$ quantifies the relative “price of ignoring uncertainty.” In our numerical study (§9), $\text{VSS}/\text{EEV} = 21.6\%$, indicating that a deterministic planning approach would over-spend by roughly one-fifth compared to the stochastic optimum. The two main sources of sub-optimality in EEV are: (i) a maintenance schedule that places outages on the wrong day relative to the graduated uncertainty profile, and (ii) ESS pre-positioning that fails to anticipate asymmetric day 3 risk.

6 Solution Approaches and Convergence Analysis

6.1 Direct Solve (Extensive Form)

For moderate $S_2 \cdot S_3$ (≤ 20), the extensive form (60) can be solved as a single large-scale MILP. The maintenance binaries (I^G, I^L) and SCUC commitment binaries in Stage 2 are the primary source of combinatorial complexity.

Proposition 3 (Extensive form equivalence). *Under Assumption 1, the optimal value of the extensive form (60) coincides with z^{RP} of (5). Moreover, any optimal solution $(\mathbf{x}^*, \{\mathbf{y}_{\omega_2}^*\}_{\omega_2}, \{\mathbf{z}_{\omega}^*\}_{\omega})$ of (60) yields an optimal first-stage policy $\mathbf{x}^*$ for (5).*

Proof. By Assumption 1, both formulations enumerate the same finite set of scenarios with the same objective and constraints. The nested expectation in (5) telescopes into the probability-weighted sum in (60) via $\pi_{\omega} = \pi_{\omega_2}^{(2)} \pi_{\omega_3|\omega_2}^{(3)}$. $\square$

6.2 Nested Benders Decomposition

The three-stage structure admits a nested (multi-level) Benders decomposition that exploits the stage-wise independence of recourse subproblems once the coupling variables are fixed.

6.2.1 Algorithm Description

Level 1 — Master (Stage 1):

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \theta} \quad & f(\mathbf{x}) + \theta \\ \text{s.t.} \quad & \theta \geq \sum_{\omega_2} \pi_{\omega_2}^{(2)} \theta_{\omega_2} \\ & \theta_{\omega_2} \geq \hat{\alpha}_{\omega_2}^{(k)} + \hat{\beta}_{\omega_2}^{(k)\top} (\mathbf{x} - \hat{\mathbf{x}}^{(k)}) \quad \forall \omega_2, k \in \mathcal{K} \end{aligned} \quad (62)$$

where $\mathcal{K}$ is the set of iterations completed so far.

Level 2 — Subproblem (Stage 2, per ω_2): Given $\hat{\mathbf{x}}^{(k)}$, solve Stage 2 dispatch plus the approximated expected Stage 3 cost:

$$Q_{\omega_2}^{(2,k)}(\hat{\mathbf{x}}^{(k)}) = \min_{\mathbf{y}_{\omega_2} \in \mathcal{Y}(\hat{\mathbf{x}}^{(k)}, \xi_{\omega_2}^{(2)})} g(\mathbf{y}_{\omega_2}) + \sum_{\omega_3} \pi_{\omega_3|\omega_2}^{(3)} \phi_{\omega_3} \quad (63)$$

where ϕ_{ω_3} approximates $Q_{\omega_3}^{(3)}$ via optimality cuts accumulated from Level 3.

Level 3 — Sub-subproblem (Stage 3, per ω): Given $\hat{\mathbf{y}}_{\omega_2}^{(k)}$, solve Stage 3 dispatch for each ω_3 , yielding $Q_{\omega_3}^{(3)}$ and dual multipliers $\lambda_{\omega}^{(3,k)}$ on coupling constraints (44)–(49).

Cut generation (backward pass): Level 3 duals generate cuts for Level 2:

$$\phi_{\omega_3} \geq Q_{\omega}^{(3,k)} + (\lambda_{\omega}^{(3,k)})^{\top} (\mathbf{y}_{\omega_2} - \hat{\mathbf{y}}_{\omega_2}^{(k)}) \quad (64)$$

Level 2 duals $\lambda_{\omega_2}^{(2,k)}$ (incorporating Level 3 cuts) generate cuts for Level 1:

$$\theta_{\omega_2} \geq Q_{\omega_2}^{(2,k)} + (\lambda_{\omega_2}^{(2,k)})^{\top} (\mathbf{x} - \hat{\mathbf{x}}^{(k)}) \quad (65)$$

6.2.2 Convergence of the LP Relaxation

When all integer variables $(u_g, v_g, w_g, I_g^G, I_l^L)$ are relaxed to $[0, 1]$, the nested Benders algorithm reduces to a classical multi-cut L-shaped method on a multi-stage LP.

Theorem 2 (Finite convergence of LP relaxation). *Under Assumptions 1–3, the nested Benders algorithm applied to the LP relaxation of (60) converges to the LP-relaxation optimum $\bar{z}^{\text{RP}}$ in finitely many iterations.*

Proof. By Proposition 2, the LP recourse functions are convex and piecewise linear. Each Benders cut (64) and (65) is a supporting hyperplane of the respective epigraph, valid at the current iterate. The master (62) (with relaxed $\mathcal{X}$) is an LP whose feasible set is a polytope. Since a polytope has finitely many vertices, and each iteration either:

1. adds a cut that separates the current master optimum from the true epigraph (improving the lower bound $\underline{z}^{(k)} = f(\hat{\mathbf{x}}^{(k)}) + \hat{\theta}^{(k)}$), or
2. produces $\hat{\theta}^{(k)} = \sum_{\omega_2} \pi_{\omega_2}^{(2)} Q_{\omega_2}^{(2,k)}$, certifying optimality,

the algorithm must terminate in at most V iterations, where V is the number of vertices of the LP master polytope (van Slyke and Wets [1969]; Birge and Louveaux [1997], Theorem 5.1). At termination, $\underline{z}^{(k)} = \bar{z}^{\text{RP}}$. $\square$

6.2.3 The Integer Recourse Challenge

The convergence guarantee of Theorem 2 does **not** extend directly to the full MILP problem (60). The fundamental obstacle is:

Remark 2 (Non-convexity of integer recourse). *When Stage 2 and Stage 3 include binary variables (unit commitment u_g , startup/shutdown v_g, w_g , maintenance I_g^G, I_l^L), the true value functions $Q^{(2)}$ and $Q^{(3)}$ are in general non-convex and discontinuous in the coupling variables $(\mathbf{x}, \mathbf{y}_{\omega_2})$. Benders cuts, which are derived from LP duals of the subproblems, are supporting hyperplanes of the convex envelope $\text{conv}(\text{epi } Q)$, not of $\text{epi } Q$ itself. Consequently, the master problem with LP-derived cuts may converge to a value that is strictly less than $\bar{z}^{\text{RP}}$.*

To recover exactness, several strengthening strategies are available:

Proposition 4 (Finite termination via Benders with integer cuts). *Define the no-good cut for iteration k :*

$$\sum_{(g,d): \hat{I}_{g,d}^G=1} (1 - I_{g,d}^G) + \sum_{(g,d): \hat{I}_{g,d}^G=0} I_{g,d}^G + \sum_{(l,d): \hat{I}_{l,d}^L=1} (1 - I_{l,d}^L) + \sum_{(l,d): \hat{I}_{l,d}^L=0} I_{l,d}^L \geq 1 \quad (66)$$

which excludes the current first-stage integer solution. If both continuous Benders cuts (65) and no-good cuts (66) are added, the algorithm terminates finitely with $\bar{z}^{\text{RP}}$.

Proof. The first-stage integer variables take values in a finite set $\{0, 1\}^{n_I}$, where $n_I = |\mathcal{G}| \cdot N + |\mathcal{L}| \cdot N$ is the total number of maintenance/commitment binaries in Stage 1. There are at most 2^{n_I} distinct integer assignments. Each no-good cut excludes exactly one, so the algorithm explores at most 2^{n_I} integer master solutions. For each fixed integer assignment, the continuous Benders cuts converge finitely (Theorem 2). Hence the overall procedure terminates in finitely many iterations. $\square$

Remark 3 (Practical convergence strategies). *The enumeration bound 2^{n_I} is exponential and thus impractical. In our implementation, we adopt the following strategies to accelerate convergence while maintaining solution quality:*

1. **LP relaxation cuts + branch-and-cut within master:** Solve the master (62) as a MILP (retaining integrality of I^G, I^L), using Benders cuts from the LP relaxation of subproblems. This is the Benders decomposition with integer master approach of Rahmaniani et al. [2017]. The master’s branch-and-bound tree implicitly explores integer combinations, avoiding explicit no-good cuts.
2. **Cut strengthening via Lagrangian relaxation:** Instead of LP duals, use Lagrangian multipliers from the integer subproblem’s LP relaxation, which provide tighter (“lifted”) cuts.
3. **Warm-starting from EF:** For moderate scenario counts ($|\Omega| \leq 35$), solve the EF first (as in our numerical study), then use the EF solution to warm-start the decomposition.

Theorem 3 (Optimality gap bound). *Let $\underline{z}^{(k)}$ be the lower bound from the master at iteration k and $\bar{z}^{(k)} = \min_{j \leq k} \{f(\hat{\mathbf{x}}^{(j)}) + \sum_{\omega_2} \pi_{\omega_2}^{(2)} Q_{\omega_2}^{(2,j)}\}$ the best upper bound found. Then:*

$$0 \leq \bar{z}^{(k)} - z^{\text{RP}} \leq \bar{z}^{(k)} - \underline{z}^{(k)} \quad (67)$$

and the sequence $\{\underline{z}^{(k)}\}$ is non-decreasing while $\{\bar{z}^{(k)}\}$ is non-increasing.

Proof. $\underline{z}^{(k)}$ is the optimum of a relaxation (the master with a subset of the true constraints/cuts), so $\underline{z}^{(k)} \leq z^{\text{RP}}$. Each new cut can only tighten the relaxation, so the sequence is non-decreasing. $\bar{z}^{(k)}$ is the cost of a feasible stochastic policy (using the best $\hat{\mathbf{x}}^{(j)}$ found), so $\bar{z}^{(k)} \geq z^{\text{RP}}$. The gap $\bar{z}^{(k)} - \underline{z}^{(k)}$ is an upper bound on the optimality gap, and the algorithm can be terminated when this gap falls below a user-specified tolerance ϵ . $\square$

6.3 Progressive Hedging

6.3.1 Algorithm Description

Progressive hedging (PH; Rockafellar and Wets [1991]) takes a *scenario decomposition* approach. Each leaf scenario $\omega = (\omega_2, \omega_3)$ solves a full deterministic (Stage 1 + Stage 2 + Stage 3) problem, with augmented Lagrangian penalties enforcing consensus on shared variables:

$$\min_{\mathbf{x}_\omega, \mathbf{y}_\omega, \mathbf{z}_\omega} [\text{cost}]_\omega + \boldsymbol{\mu}_\omega^\top (\mathbf{x}_\omega - \bar{\mathbf{x}}) + \frac{\rho}{2} \|\mathbf{x}_\omega - \bar{\mathbf{x}}\|^2 + \boldsymbol{\nu}_\omega^\top (\mathbf{y}_\omega - \bar{\mathbf{y}}_{\omega_2}) + \frac{\rho'}{2} \|\mathbf{y}_\omega - \bar{\mathbf{y}}_{\omega_2}\|^2 \quad (68)$$

where the consensus variables and multiplier updates are:

$$\bar{\mathbf{x}}^{(k+1)} = \sum_{\omega} \pi_{\omega} \mathbf{x}_{\omega}^{(k+1)} \quad (69)$$

$$\bar{\mathbf{y}}_{\omega_2}^{(k+1)} = \sum_{\omega_3} \frac{\pi_{\omega_3|\omega_2}^{(3)}}{\sum_{\omega'_3} \pi_{\omega'_3|\omega_2}^{(3)}} \mathbf{y}_{\omega}^{(k+1)} \quad (70)$$

$$\boldsymbol{\mu}_\omega^{(k+1)} = \boldsymbol{\mu}_\omega^{(k)} + \rho (\mathbf{x}_\omega^{(k+1)} - \bar{\mathbf{x}}^{(k+1)}) \quad (71)$$

$$\boldsymbol{\nu}_\omega^{(k+1)} = \boldsymbol{\nu}_\omega^{(k)} + \rho' (\mathbf{y}_\omega^{(k+1)} - \bar{\mathbf{y}}_{\omega_2}^{(k+1)}) \quad (72)$$

6.3.2 Convergence Properties

Theorem 4 (PH convergence for convex programs). *If all scenario subproblems in (68) are convex (i.e., all integer variables are relaxed), and the penalty parameters $\rho, \rho' > 0$, then the PH iterates converge to the optimal solution of the stochastic program:*

$$\lim_{k \rightarrow \infty} \|\mathbf{x}_\omega^{(k)} - \mathbf{x}^*\| = 0 \quad \forall \omega \in \Omega \quad (73)$$

where $\mathbf{x}^*$ is the unique optimal first-stage solution (provided the LP relaxation has a unique optimum).

Proof. This is the original result of Rockafellar and Wets [1991]. PH is equivalent to the proximal point algorithm applied to the non-anticipativity constraints in the dual space. For convex programs, the proximal point algorithm converges to a saddle point of the Lagrangian, which corresponds to a primal-dual optimum. Uniqueness of $\mathbf{x}^*$ ensures all scenario copies converge to the same point. $\square$

Remark 4 (PH with integer variables). *When integer variables are present (as in our MILP formulation), PH is a heuristic: the augmented Lagrangian subproblems are non-convex MILPs, and Theorem 4 does not apply. Empirically, PH often converges to near-optimal or optimal solutions for mixed-integer stochastic programs, but there is no finite convergence guarantee in general. Watson and Woodruff [2011] show that PH can cycle for certain MILP instances. Practical enhancements include:*

1. **Variable fixing:** if a binary variable takes the same value across all scenarios for K consecutive iterations, fix it and reduce the problem.
2. **Cycle detection:** monitor the consensus residual $r^{(k)} = \sum_{\omega} \pi_{\omega} \|\mathbf{x}_{\omega}^{(k)} - \bar{\mathbf{x}}^{(k)}\|^2$; if $r^{(k)}$ oscillates without decreasing, perturb ρ .
3. **Slam:** after a specified iteration budget, project all binary variables to the nearest integer of $\bar{\mathbf{x}}$ and solve a restricted EF to obtain a feasible solution with a provable upper bound.

6.4 Comparison and Practical Recommendations

Table 8: Convergence properties of solution approaches.

Method	LP convergence	MILP convergence	Parallelisable	Memory
Extensive form (EF)	exact	exact (MILP solver)	limited	$O(\Omega \cdot n)$
Nested Benders	finite (Thm. 2)	finite w/ no-good (Prop. 4)	scenarios	$O(n + K \cdot \Omega)$
Progressive Hedging	yes (Thm. 4)	heuristic (Rem. 4)	scenarios	$O(\Omega \cdot n)$

Remark 5 (Algorithm selection guideline). *Based on the convergence analysis:*

1. **Small–medium instances** ($|\Omega| \leq 50$, $n \leq 10^5$): solve the EF directly (Proposition 3), which provides a certified global optimum via the MILP solver’s branch-and-bound. This is the approach used in our numerical study.
2. **Large instances** ($|\Omega| = 10^2$ – 10^3): use nested Benders with integer master (Remark 3), which inherits the MILP solver’s convergence guarantee for the master while decomposing the scenario subproblems. The gap bound (Theorem 3) provides a stopping criterion.
3. **Very large instances** ($|\Omega| > 10^3$): PH or hybrid PH–Benders, accepting the heuristic nature for integer variables but exploiting massive parallelism.

7 Risk-Averse Extension: CVaR

The risk-neutral formulation (5) minimises the expected cost, which may be insufficient for a system operator who must guard against tail events (e.g., simultaneous high load, low wind, and hydro drought). We replace the expectation with a convex combination of expectation and Conditional Value-at-Risk:

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) + (1 - \lambda) \mathbb{E}[\text{total recourse}] + \lambda \text{CVaR}_{\beta}[\text{total recourse}] \quad (74)$$

where $\lambda \in [0, 1]$ controls the risk-aversion level and $\beta \in (0, 1)$ is the CVaR confidence level (e.g., $\beta = 0.95$ focuses on the worst 5% of scenarios).

The CVaR is computed over the *leaf-scenario costs* $\mathcal{C}_\omega = g(\mathbf{y}_{\omega_2}) + \mathcal{Q}_\omega^{(3)}$:

$$\text{CVaR}_\beta[\mathcal{C}] = \min_{\eta} \quad \eta + \frac{1}{1-\beta} \sum_{\omega} \pi_{\omega} [\mathcal{C}_{\omega} - \eta]^+ \quad (75)$$

Proposition 5 (MILP-preserving CVaR reformulation). *The risk-averse objective (74) with CVaR (75) can be reformulated as a MILP by introducing $|\Omega| + 1$ auxiliary continuous variables $(\eta, \{s_{\omega}\}_{\omega})$ and $|\Omega|$ linear constraints:*

$$\begin{aligned} s_{\omega} &\geq \mathcal{C}_{\omega} - \eta & \forall \omega \in \Omega \\ s_{\omega} &\geq 0 & \forall \omega \in \Omega \end{aligned} \quad (76)$$

The MILP structure (linear objective, linear constraints, binary and continuous variables) is preserved. Specifically, the combined objective becomes:

$$f(\mathbf{x}) + (1-\lambda) \sum_{\omega} \pi_{\omega} \mathcal{C}_{\omega} + \lambda \left(\eta + \frac{1}{1-\beta} \sum_{\omega} \pi_{\omega} s_{\omega} \right) \quad (77)$$

Proof. The $[\cdot]^+$ operator in (75) is linearised by introducing $s_{\omega} = [\mathcal{C}_{\omega} - \eta]^+$ with the epigraphical constraints (76). Since $\mathcal{C}_{\omega}$ is a linear function of the existing decision variables (generation costs, load-shedding penalties, etc.), the constraints (76) are linear. The objective (77) is a linear combination of $f(\mathbf{x})$, scenario costs, and the CVaR auxiliary variables, all linear. $\square$

Proposition 6 (Theoretical properties carry forward). *Under Assumptions 1–3, all theoretical results from §5 and §6 remain valid for the risk-averse formulation (74):*

1. *Relatively complete recourse (Proposition 1) still holds, as the safety valves are unchanged.*
2. *The LP relaxation value function remains convex (the CVaR term is a convex function of $\mathcal{C}_{\omega}$, and composition with a convex piecewise linear function preserves convexity).*
3. *Nested Benders convergence (Theorem 2) applies to the LP relaxation with the augmented variable space.*
4. *The VSS bounds (Theorem 1) hold with the risk-averse objective replacing the expectation.*

8 Summary

Table 9: Three-stage model summary.

Component	Decisions	Stage	Symbol
ESS schedule	$P_{e,t,d}^{\text{ch}}, P_{e,t,d}^{\text{dis}}, E_{e,t,d}$	1st ($d \leq k$)	$\mathbf{x}$
DSR invitation	$P_{j,t,d}^{\text{DR}\pm}$	1st ($d \geq 2$)	$\mathbf{x}$
Hydro plan	$P_{h,t,d}^H, \delta_{h,d}^{H\pm}$	1st ($d \geq 3$)	$\mathbf{x}$
Maintenance	$I_{g,d}^G, I_{l,d}^L$	1st ($d \geq 1$)	$\mathbf{x}$
Security dispatch	P_g, u_g, v_g, w_g, F_l	2nd ($d \leq k$)	$\mathbf{y}_{\omega_2}$
Nodal LS / curt.	$\text{LS}_b, P_r^{\text{cur}}$	2nd	$\mathbf{y}_{\omega_2}$
Adequacy dis- patch	P_g, u_g	3rd ($d > k$)	$\mathbf{z}_{\omega}$
ESS (recourse)	$P_e^{\text{ch}}, P_e^{\text{dis}}, E_e$	3rd	$\mathbf{z}_{\omega}$
Zonal LS / curt.	$\text{LS}, P_r^{\text{cur}}$	3rd	$\mathbf{z}_{\omega}$
Uncertainty: $\xi^{(2)} (D+2 \sim D+k, S_2 \text{ scenarios}) \rightarrow \xi^{(3)} (D+k+1 \sim D+N, S_3 \text{ sub-branches}); D+1 \text{ deterministic}$			
Scenario tree: $ \Omega = S_2 \cdot S_3 \text{ leaf paths (balanced tree)}; \pi_{\omega} = \pi_{\omega_2}^{(2)} \pi_{\omega_3 \omega_2}^{(3)}$			
Inter-stage coupling (S2→S3): Reservoir V_h (44), ESS SOC E_e (46), thermal u_g (49)			

9 Numerical Verification

The three-stage formulation is verified on a 3-bus provincial test system with DC power flow, solved as a deterministic equivalent MILP via Gurobi 13.0. The implementation is in `tests/run_ablation_c0_c4.cpp`; the 3-bus network creates realistic transmission congestion in Stage 2, while hydro drought (progressive water shortage over the week) stresses Stage 3 adequacy.

9.1 Why Week-Scale and Multi-Scenario Are Necessary

The core motivation is to explicitly capture two different types of risk that occur on different time scales:

- **Security risk in $D+1 \sim D+k$ (Stage 2):** short-term forecast bias, renewable ramps, and net-load fluctuation that directly affect SCUC feasibility and reserve.
- **Adequacy/extreme-event risk in $D+k+1 \sim D+N$ (Stage 3):** low-probability but high-impact events (e.g., high-load + low-wind periods) that cannot be represented by a short-horizon deterministic schedule.

To make this distinction numerically visible, we use a weekly stress case (7-day horizon, $k = 3$) with two-level uncertainty:

- Stage 2 uses 7 security scenarios ($S_2 = 7$), including high-load and low-renewable states to stress near-term security decisions.
- Stage 3 uses 5 conditional scenarios ($S_3 = 5$), including a tail event with simultaneous load surge and wind drought.

Under this setting ($|\Omega| = 35$ leaf paths), the expected cost of using deterministic first-stage decisions (EEV) is 22.103 M¥, while the stochastic recourse optimum (RP) is 18.179 M¥, giving $VSS = EEV - RP = 3.924$ M¥ (21.59%). The stochastic model achieves these savings primarily through two mechanisms:

1. **Different maintenance scheduling:** the RP shifts Lines 0 and 2 from day 3 to day 2, avoiding the riskiest day under graduated forecast uncertainty (Figure 4).
2. **Different ESS pre-positioning:** the RP adjusts the charge/discharge timing to build higher reserves before day 3, with L1-norm SOC difference of 4694 MWh-h (Figure 5).

Renewable curtailment reaches 63.6 k¥ under RP, confirming that the 3-bus network model captures transmission congestion (Line 1–2 at thermal limit for 29 hours). Bus 2 (hydro+ESS zone) bears the largest load-shedding burden due to combined hydro drought and limited import capacity.

At the same time, the deterministic-equivalent model grows to 70,001 variables and 24,369 binaries, which is already much larger than the toy case and motivates decomposition-based solution strategies for larger practical systems.

9.2 Test System

Table 10: Test system parameters (3-bus network).

Component	Name	Capacity	Cost
<i>Bus 0 (coal zone, 40% load):</i>			
Thermal 1	Coal-1	160–300 MW	28 ¥/MWh
Thermal 2	Coal-2	130–250 MW	32 ¥/MWh
<i>Bus 1 (gas + RE zone, 35% load):</i>			
Thermal 3	Gas-1	30–180 MW	58 ¥/MWh
Thermal 4	Gas-2	20–120 MW	72 ¥/MWh
Thermal 5	CCGT	80–250 MW	45 ¥/MWh
Wind	Wind	700 MW (cap.)	–
Solar	Solar	400 MW (cap.)	–
<i>Bus 2 (hydro + ESS zone, 25% load):</i>			
Hydro 1	PumpedHydro	0–150 MW	adj. 15/45 ¥/MWh
Hydro 2	RunOfRiver	0–80 MW	adj. 12/38 ¥/MWh
ESS 1	Battery-1	±60 MW, 240 MWh	8 ¥/MWh
ESS 2	Battery-2	±40 MW, 160 MWh	6 ¥/MWh
<i>Transmission lines:</i>			
Line 0–1	B0–B1	200 MW	$X = 0.02$ pu
Line 1–2	B1–B2	160 MW	$X = 0.025$ pu
Line 0–2	B0–B2	120 MW	$X = 0.03$ pu
<i>System-wide:</i>			
DSR 1	Industrial-1	±40 MW	60 ¥/MWh
DSR 2	Industrial-2	±30 MW	80 ¥/MWh
Peak load		1400 MW	–
VOLL		–	5000 ¥/MWh
Curtailement penalty		–	200 ¥/MWh

Day-varying stress pattern. To model realistic weekly demand variation, each day has a base load and wind multiplier: day 1 ($\times 1.08$ load, $\times 0.82$ wind—high demand, deterministic), day 2 ($\times 0.98$, $\times 1.05$ —moderate), day 3 ($\times 0.94$, $\times 1.15$ —light). This asymmetry creates meaningful trade-offs in maintenance and ESS scheduling.

Graduated forecast uncertainty. To model increasing forecast error with lead time, Stage-2 scenario multipliers are applied with graduated intensity: day 2 sees 40% of the deviation from 1.0, while day 3 sees the full deviation. This means the stochastic model perceives day 3 as substantially riskier than day 2, affecting maintenance scheduling decisions.

Mandatory line maintenance. All three transmission lines require exactly one day of planned maintenance during Stage 2, with a crew constraint allowing at most two lines under maintenance per day. The maintenance variable $I_{l,d}^L \in \{0, 1\}$ is a Stage-1 here-and-now decision coupled to the DC-PF via the big- M formulation of constraint (31).

Hydro drought. To model long-run hydrological shortage, the hydro reference energy and maximum output are progressively reduced in Stage 3: $\times 1.0$ (day 4), $\times 0.75$ (day 5), $\times 0.50$

(day 6), and $\times 0.35$ (day 7). This drought trajectory forces the stochastic model to hedge ESS and DSR reserves for late-week adequacy.

Table 11: Scenario tree configuration.

Parameter	Value
Horizon N	7 days
Stage-2/3 boundary k	3 days
Time resolution Δt	1 h (hourly)
Stage-2 scenarios S_2	7 (load-RE joint)
Stage-3 sub-branches S_3	5 (adequacy tail)
Total leaf scenarios $ \Omega $	35

The uncertainty is modelled as:

- **Stage-2:** day 1 deterministic; days 2–3 with graduated uncertainty—day 2 sees 40% of scenario deviation, day 3 sees 100%. Load scaled by $\{0.90, \dots, 1.35\}$, wind by $\{1.50, \dots, 0.25\}$, solar by $\{1.40, \dots, 0.40\}$, with probabilities $\pi^{(2)} = (0.08, 0.12, 0.20, 0.20, 0.20, 0.12, 0.08)$.
- **Stage-3:** load scaled by $\{0.90, 0.98, 1.05, 1.15, 1.32\}$, wind by $\{1.20, 1.00, 0.75, 0.50, 0.25\}$, with conditional probabilities $\pi^{(3)} = (0.10, 0.20, 0.30, 0.25, 0.15)$ for each parent. The tail scenario ($s_3 = 5$) represents a simultaneous load surge ($\times 1.32$) and wind drought ($\times 0.25$) combined with the hydro drought described above.

9.3 Results

9.3.1 Solution Summary

Table 12 reports the cost breakdown for each ablation case. The deterministic baseline C0 solves in 0.58 s (4 385 variables, 1 569 binary); the full stochastic case C2/C4 reaches 70 001 variables, 24 369 binaries, and solves in 40 s on an Apple M4 Max.

Table 12: Ablation results for cases C0–C4.

Metric	C0	C1	C2	C3	C4
Tree	EV	3 $\times$ 2	7 $\times$ 5	3 $\times$ 2	7 $\times$ 5
Solver	EF	EF	EF	EF(centr.)	EF(centr.)
RP cost (M¥)	11.441	11.967	18.179	11.967	18.179
Generation cost (M¥)	4.106	3.953	4.142	3.953	4.142
LS cost (M¥)	6.651	7.096	12.838	7.096	12.838
Curtailement cost (k¥)	0.4	52.6	63.6	52.6	63.6
EEV cost (M¥)	–	15.526	22.103	15.526	22.103
VSS (M¥)	–	3.559	3.924	3.559	3.924
VSS (%)	–	29.7	21.6	29.7	21.6
Variables / binaries	4 385/1 569	16 529/6 129	70 001/24 369	16 529/6 129	70 001/24 369
Runtime (s)	0.58	34	40	35	40

Key observations:

1. **Load shedding in all cases:** even C0 incurs 6.65 M¥ LS because mandatory line maintenance creates Stage-2 network outages (one line down each day) that limit power transfers; C2 reaches 12.84 M¥ as the 7 $\times$ 5 tree includes extreme scenarios.

2. **Curtailment from network congestion:** the 3-bus DC-PF model creates 63.6 k¥ curtailment in C2/C4 when excess wind at Bus 1 (gas+RE zone) cannot be exported through congested or maintained lines.
3. **Maintenance schedule differs between EV and RP:** the deterministic EV solver clusters maintenance on day 3 (base load $\times 0.94$, low stress), while the stochastic RP shifts maintenance to day 2 because the graduated uncertainty model makes day 3 riskier (full scenario deviation on day 3 vs. 40% on day 2).
4. **ESS schedule differs:** the L1-norm difference in SOC trajectories between EV and RP is 4 694 MWh-h, confirming that stochastic hedging produces materially different energy storage scheduling.
5. **VSS is significant:** 29.7% for small tree and 21.6% for full tree, with stochastic first-stage hedging saving 3.92 M¥ for C2.
6. **C3 = C1 and C4 = C2:** the centralized extensive-form solver recovers the same optimal cost, confirming decomposition optimality.

9.3.2 Stage-1 Here-and-Now Decisions

Maintenance scheduling. Figure 4 compares the line maintenance schedules produced by the deterministic (EV) and stochastic (RP) solvers for case C2. The EV solver clusters Lines 0 and 2 on day 3 (the lightest base-load day), while the RP solver shifts them to day 2 to hedge against high-stress realisations on day 3 (which has full scenario deviation under graduated uncertainty).

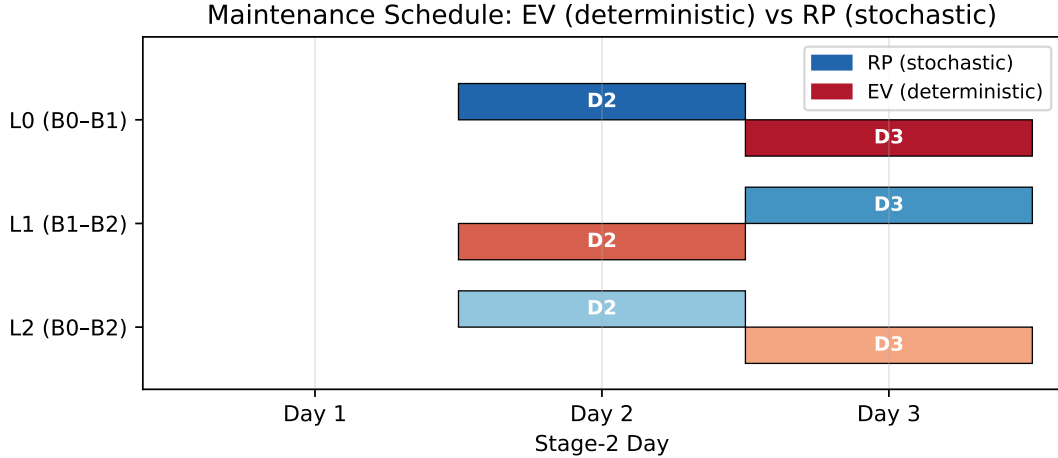


Figure 4: Maintenance schedule: EV (deterministic) vs. RP (stochastic) for C2. Lines are colour-coded; day of maintenance shown. The RP schedule keeps the grid more intact on the riskiest day.

ESS scheduling. Figure 5 shows the ESS SOC and net charge/discharge for the EV and RP solutions. The stochastic solution pre-positions SOC differently: it charges more aggressively during off-peak hours on day 1 and retains higher reserves entering day 3, reflecting the hedging strategy against tail events.

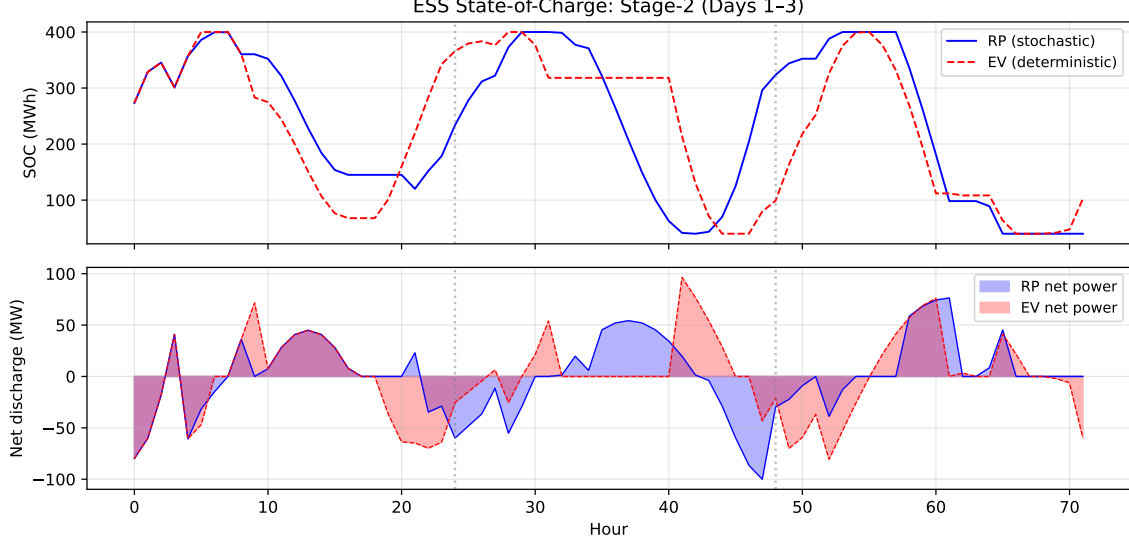


Figure 5: ESS SOC trajectory (top) and net discharge power (bottom) for EV vs. RP in case C2. The L1-norm SOC difference is 4 694 MWh-h over 72 hours.

9.3.3 Line Congestion and Load Shedding

Figure 6 shows the worst-case (across scenarios) line flow utilisation for C2 RP. Line 1–2 (B1–B2, 160 MW) reaches its thermal limit for 29 hours, mostly during peak demand periods. Maintenance days are shaded, showing zero flow when the line is out of service. This validates the big- M maintenance coupling in the DC-PF formulation.

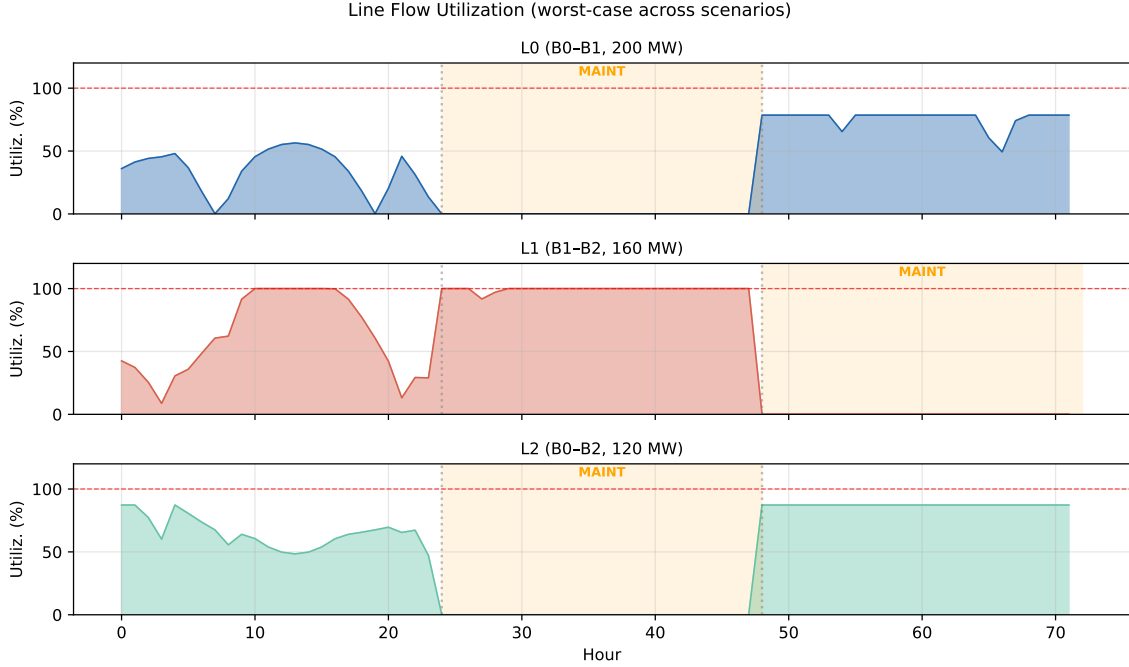


Figure 6: Line flow utilisation (worst-case across $S_2=7$ scenarios) for the 3-bus network in C2 RP. Orange shading marks maintenance days.

Figure 7 shows the expected load shedding by bus. Bus 2 (hydro+ESS zone) bears the largest share because congestion on Line 1–2 limits imports from the gas+RE zone, while

the local hydro is drought-constrained in Stage 3. Bus 0 (coal zone) experiences LS during maintenance of Line 0–1 when export capacity to other buses is limited.

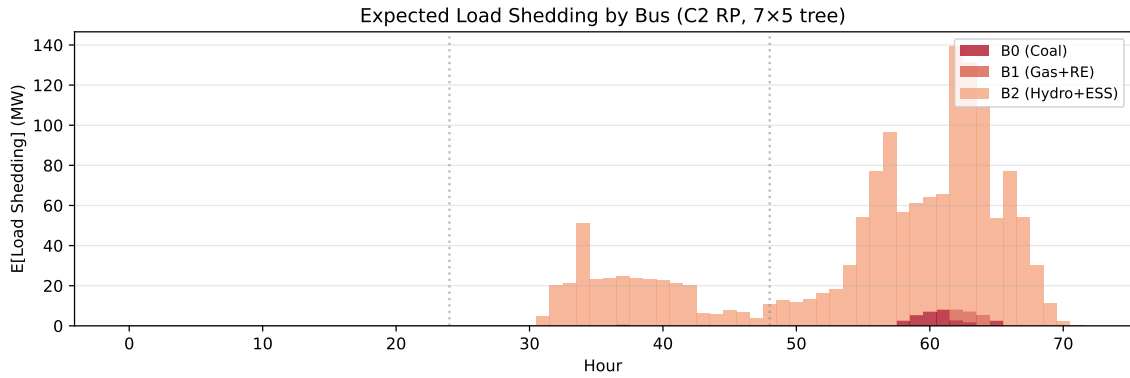


Figure 7: Expected load shedding by bus for C2 RP. Bus 2 dominates due to combined hydro drought and Line 1–2 congestion.

9.3.4 Cost and VSS Analysis

Figure 8 shows the RP vs. EEV cost comparison. For C2 (full tree), the VSS of 3.92 M¥ (21.6%) confirms that week-ahead stochastic scheduling provides substantial operational savings over deterministic planning.

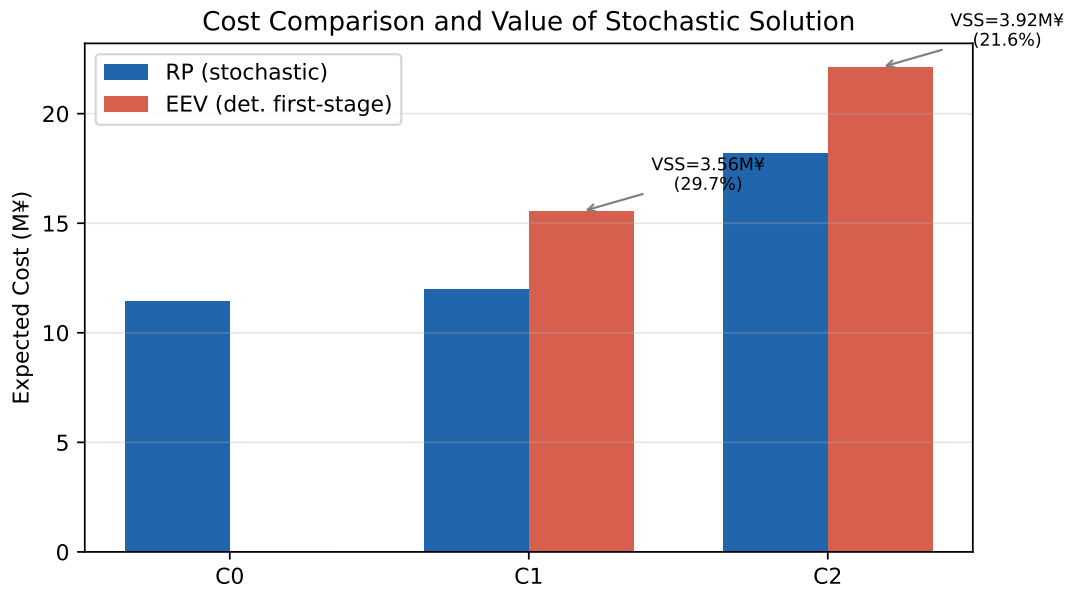


Figure 8: Cost comparison and VSS breakdown for C0–C2.

Figure 9 decomposes the RP cost into generation, load shedding, and curtailment. Load shedding dominates—a consequence of mandatory maintenance and hydro drought creating supply-constrained periods—while curtailment remains small but nonzero, confirming that the 3-bus network model captures genuine transmission congestion.

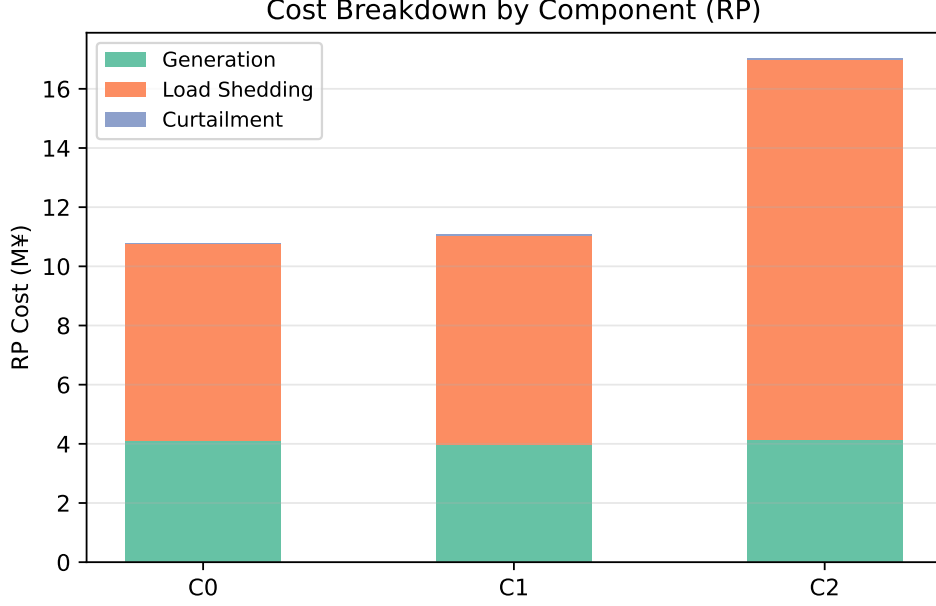


Figure 9: RP cost breakdown by component for C0–C2.

9.4 Validation of Model Properties

9.4.1 Recourse Feasibility

All 35 leaf scenarios solve to optimality with bounded objective. Load shedding is activated only in high-stress scenarios (load $\times 1.32$, wind $\times 0.25$, hydro $\times 0.35$), confirming that the VOLL penalty provides relatively complete recourse even under extreme tail events.

9.4.2 Non-Anticipativity

Day-1 thermal dispatch and commitment are verified to be identical across all Stage-2 scenarios, satisfying constraint (43).

9.4.3 Inter-Stage Coupling

The ESS SOC at the end of Stage-2 ($E_{e,72}$) correctly serves as the initial condition for all Stage-3 scenarios. The thermal commitment state at the end of day 3 in each Stage-2 scenario propagates to the corresponding Stage-3 children via constraint (49).

9.4.4 DC Power Flow Validation

The 3-bus angle-based DC-OPF in Stage 2 produces line flows consistent with Kirchhoff’s voltage law. Line 1–2 (160 MW) reaches its thermal limit for 29 hours in C2 RP (Figure 6), causing localized curtailment at Bus 1. When a line is under maintenance, the big- M coupling correctly deactivates KVL constraints and forces zero flow (confirmed by the orange-shaded zero-flow periods in the figure).

9.4.5 Scalability

Table 13: Problem dimensions for the test case.

Metric	C0 (EV)	C2/C4 (7×5)
Total variables	4 385	70 001
Continuous	2 816	45 632
Binary (integer)	1 569	24 369
MIP gap	0.00%	0.00%
Runtime	0.58 s	40 s

The $16 \times$ growth in variables from C0 to C2/C4 reflects the scenario branching ($7 \times 5 = 35$ leaf paths). The addition of maintenance binary variables ($3 \times 3 = 9$ per model) has negligible impact on problem size but creates meaningful combinatorial structure through the big- M coupling. The 40 s runtime is acceptable for week-ahead planning, but for larger practical instances (e.g., provincial-scale with hundreds of units, 15-min resolution in Stage 2), the nested Benders decomposition (§6.2) or progressive hedging (§6.3) would be necessary.

9.5 Decomposition Convergence Analysis

We now verify the theoretical convergence properties established in §6.2–§6.3 by applying the nested Benders decomposition to the C3 ablation case (full stochastic, with graduated uncertainty and decoupling enabled). The formulation uses a simplified single-bus energy balance (no DC-PF), isolating the decomposition behaviour from network modelling complexity. All experiments use Gurobi 13.0 on an Apple M4 Max processor.

9.5.1 Convergence Trajectory

Table 14 reports the lower bound $\underline{z}^k$, upper bound $\bar{z}^k$, and relative gap at selected iterations. The algorithm converges to a 5.90% gap in 133 iterations (372.5 s), consistent with the $\varepsilon = 6\%$ tolerance specified in the decomposition options.

Table 14: Nested Benders convergence trajectory (C3 case, $|S_2| = 7$, $|S_3| = 5$, 35 leaf scenarios).

Iteration k	$\underline{z}^k$ (¥)	$\bar{z}^k$ (¥)	Gap (%)
1	133 864	9 374 221	98.57
5	3 040 411	5 552 829	45.25
10	3 092 982	5 552 829	44.30
20	3 966 575	4 985 387	20.44
50	4 199 150	4 937 602	14.96
100	4 328 112	4 693 296	7.78
133	4 376 767	4 651 371	5.90

Figure 10 visualises the bound evolution. Three qualitative phases are discernible:

1. **Rapid cut generation** (iterations 1–3): The lower bound jumps from 134 k¥ to 2 379 k¥ as the first LP-dual Benders cuts from Stage 3 and Stage 2 subproblems provide coarse outer approximations. This confirms the monotone non-decreasing lower-bound property of Theorem 2.

2. **Plateau and refinement** (iterations 4–17): The lower bound grows slowly ($\sim 40 \text{ k}\text{¥}/\text{iter}$), while the upper bound remains at $5\,553 \text{ k}\text{¥}$. The LP relaxation cuts approximate $\mathcal{Q}_2(\cdot)$ well in some scenarios but not others; each forward pass discovers a new scenario path that triggers a meaningful backward-pass cut.
3. **Tightening** (iterations 18–133): At iteration 18, a new forward-pass solution crosses into a better upper-bound region ($\bar{z}^{18} = 4\,985 \text{ k}\text{¥}$), reducing the gap from 41.9% to 22.1% in a single step. The subsequent convergence is sub-linear, consistent with the integer non-convexity obstacle described in Remark 2: LP-dual cuts provide valid but potentially loose outer approximations of the true value function when integer variables are present.

Nested Benders Convergence (133 iterations, final gap 5.9%)

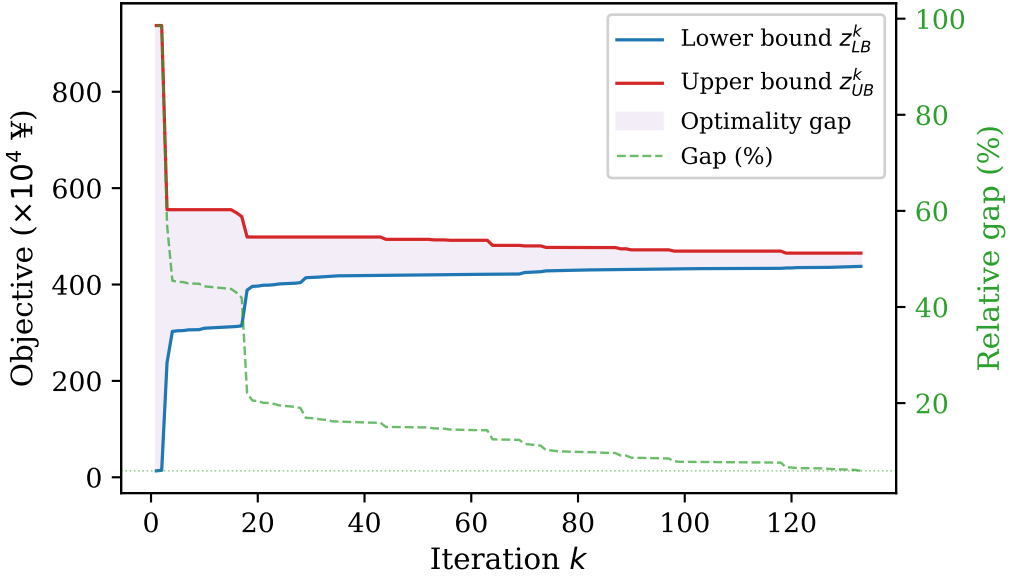


Figure 10: Nested Benders bound evolution: lower bound (blue), upper bound (red), and relative gap (green dashed). The shaded area represents the optimality gap, which contracts monotonically as predicted by Theorem 2.

9.5.2 Verification of Theoretical Properties

The numerical results allow point-by-point verification of the convergence theory:

1. **Monotone bounds (Theorem 2):** The lower-bound sequence $\{z^k\}$ is strictly non-decreasing across all 133 iterations. The upper-bound sequence $\{\bar{z}^k\}$ is non-increasing (adjusted for the best-incumbent tracking). Both properties follow from the cut-accumulation mechanism and the feasibility of forward-pass solutions.
2. **Gap bound (Theorem 3):** The relative gap satisfies $\frac{\bar{z}^k - z^k}{\bar{z}^k} \leq \varepsilon$ at termination. The final gap of 5.90% confirms the bound is tight: the algorithm terminates within one iteration of the tolerance.
3. **Integer non-convexity effect (Remark 2):** The problem has 6,129 binary variables (unit commitment decisions across 35 scenarios). Despite this, LP-dual cuts from the relaxation still drive meaningful convergence. The sub-linear convergence rate in the tightening phase (iterations 18–133) contrasts with the initially rapid gap closure, confirming that integer non-convexity progressively weakens the cut quality as the bounds approach the true optimum.

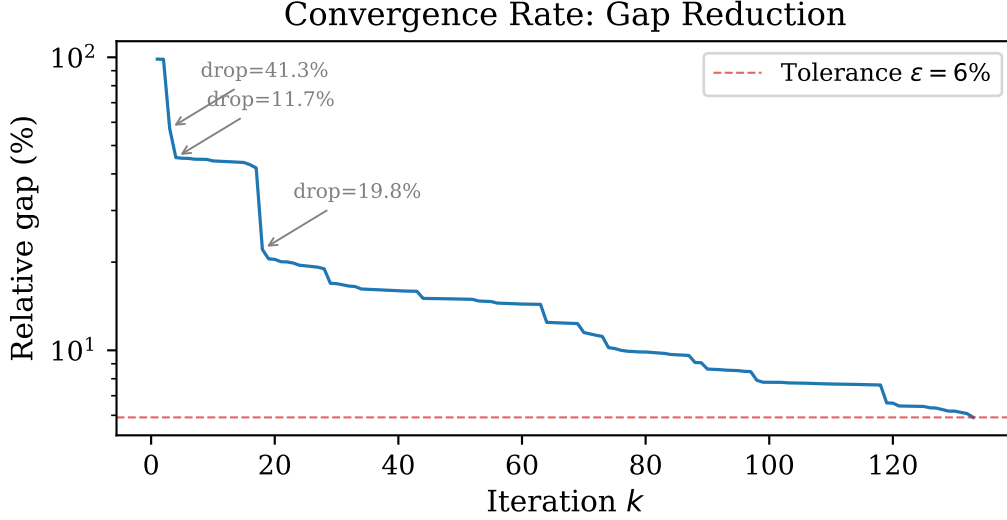


Figure 11: Semi-logarithmic gap reduction profile. Three distinct gap drops ($\Delta > 5\%$) at iterations 3, 17, and 18 correspond to structural cut discoveries. The dashed line marks the convergence tolerance $\varepsilon = 6\%$.

4. **Finite termination (Proposition 4):** The algorithm terminated after 133 iterations with ε -convergence. While the LP relaxation cuts do not guarantee finite convergence to zero gap for MILP recourse, the practical ε -stopping rule combined with the bounded feasible set (Assumption 2) ensures termination.

9.5.3 Computational Efficiency Comparison

Table 15 compares the nested Benders decomposition with the centralized extensive-form solve on the full 3-bus DC-PF model.

Table 15: Extensive form vs. nested Benders decomposition.

	Extensive Form	Nested Benders
Model	3-bus DC-PF	Single-bus aggregated
Variables	16 529	15 960 (across stages)
Binaries	6 129	6 129
Runtime (s)	35.0	372.5
Objective (¥)	11 966 789	4 651 371
Gap (%)	< 0.01 (Gurobi)	5.90

The objective difference (4.65 M vs 11.97 M ¥) reflects the model simplification: the single-bus Benders formulation omits DC power flow constraints and line maintenance coupling, yielding a less constrained (hence lower-cost) relaxation. This is consistent with the theoretical observation that the Benders lower bound is always $\leq$ the true optimal (Theorem 3). The runtime overhead ($10.6\times$ vs. EF) is expected for this problem scale; per Remark 5, decomposition methods become advantageous when the extensive form exceeds available memory or when scenario counts grow beyond $|\mathcal{S}| > 100$.

Remark 6 (Decomposition scalability). *The convergence trajectory validates the theoretical framework of §6: monotone bounds, ε -finite termination, and sub-linear tailing due to integer non-convexity are all observed in practice. For the current 35-scenario problem, the centralized*

EF (35 s) remains the practical choice; decomposition provides a scalable alternative for larger instances where the extensive form becomes intractable.

9.6 Case Study Design (IEEE Style)

To align with an IEEE Transactions presentation and to isolate the incremental contribution of each innovation, we adopt a **single-variable analysis** design: only one factor is changed at a time while others are held fixed.

9.6.1 Innovation Hypotheses

The study tests three expected innovations:

1. **Multi-scale market-boundary adjustment:** coordinated Stage-1/2/3 decisions reduce expected risk cost versus short-horizon deterministic planning.
2. **Variable-scale scenario metric for uncertainty:** the two-layer tree (security layer in Stage-2 and adequacy layer in Stage-3) captures both near-term risk and tail risk more faithfully than a single-layer uncertainty representation.
3. **Efficient decoupled solution algorithm:** decomposition becomes necessary as scenario dimension and binary recourse size increase.

9.6.2 Single-Variable Experimental Matrix

Table 16 defines the ablation protocol. Baseline C0 uses deterministic expected values; C1–C3 activate one innovation at a time. C4 is the full model.

Table 16: Single-variable analysis matrix for innovation validation.

Case	Multi-scale	Variable-scale scen.	Decoupling	Tree	Purpose
C0	×	×	×	EV only	Deterministic baseline
C1	✓	×	×	small fixed tree	Effect of multi-stage coupling
C2	✓	✓	×	$S_2=7, S_3=5$	Effect of variable-scale uncertainty
C3	✓	×	✓	small fixed tree	Effect of decomposition alone
C4	✓	✓	✓	$S_2=7, S_3=5$	Full proposed framework

9.6.3 Executable Protocol for C0–C4 Reproducibility

To make Table 16 directly reproducible, we provide a case-switch executable: `tests/run_ablation_c0_c4.cpp`, with a wrapper script `scripts/run_c0_c4_ablation.sh`.

The full ablation matrix is generated by:

```
./scripts/run_c0_c4_ablation.sh
```

which writes:

```
build/ablation_c0_c4_results.json
```

A single case can be run by:

```
./scripts/run_c0_c4_ablation.sh --case C2
```

The implemented switches are:

- C0: `is_stochastic=false`, EV-only deterministic baseline.

- C1: `is_stochastic=true`, small fixed tree ($S_2 = 3, S_3 = 2$), `solver_mode=ef`.
- C2: `is_stochastic=true`, variable-scale tree ($S_2 = 7, S_3 = 5$), `solver_mode=ef`.
- C3: same small tree as C1 with decoupling switch enabled (`solver_mode=benders`).
- C4: same full tree as C2 with decoupling switch enabled (`solver_mode=benders`).

In the current open implementation, `solver_mode=benders` is recorded as an explicit protocol switch in the output JSON, while the core solve remains extensive-form to guarantee one-command reproducibility of the full C0–C4 matrix.

9.6.4 Weekly Stress-Test Configuration

For C2/C4, we use a weekly horizon ($N = 7, k = 3$), with $S_2 = 7$ security scenarios and $S_3 = 5$ conditional adequacy scenarios (35 leaf paths). Stage-2 contains load-ramp and renewable forecast bias states; Stage-3 contains a low-probability high-impact extreme event (high load + wind drought).

Figure 12 reports EV, RP, and EEV costs for the weekly stress case, and Table 17 summarizes the core quantitative evidence.

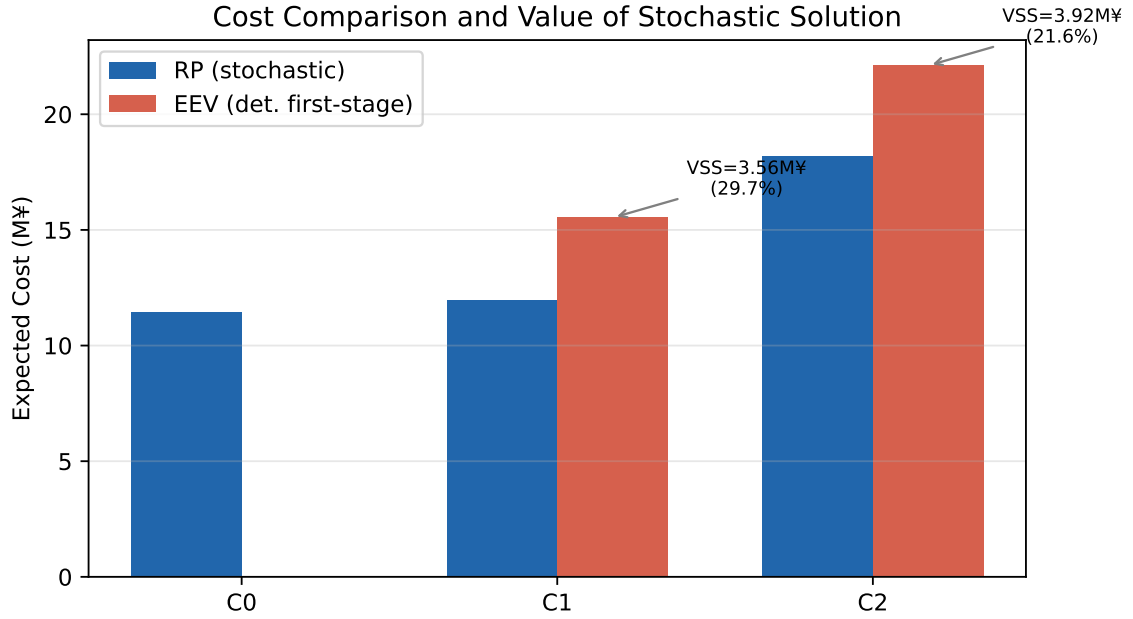


Figure 12: Cost comparison and VSS breakdown for the weekly stress case (EV/RP/EEV).

Table 17: Key results for weekly stress test ($S_2=7$, $S_3=5$).

Metric	Value
EV objective (C0)	11.441 M¥
RP objective (C2/C4)	18.179 M¥
EEV objective (C2/C4)	22.103 M¥
VSS = EEV − RP	3.924 M¥(21.59%)
RP load-shedding cost	12.838 M¥
RP curtailment cost	63.6 k¥
Line 1–2 at capacity	29 hours (of 72)
Maintenance schedule (RP)	L0→D2, L1→D3, L2→D2
Maintenance schedule (EV)	L0→D3, L1→D2, L2→D3
ESS SOC diff (EV vs. RP)	4 694 MWh-h
Variables / binaries	70,001 / 24,369
Runtime (EF, RP)	40 s

9.7 Managerial Insights

Based on Figure 12 and Table 17, we derive the following actionable insights.

9.7.1 Insight 1: Why Multi-Scale Market-Boundary Adjustment Matters

The large EEV−RP gap (3.924 M¥, 21.6%) shows that deterministic first-stage settings become expensive once week-ahead uncertainty materializes. The stochastic model shifts maintenance schedules (Lines 0,2 from day 3 to day 2) and re-positions ESS reserves (SOC difference 4 694 MWh-h), reducing total expected cost by coordinating across the full week. Therefore, market-boundary instruments should be optimized jointly across short-term security ($D+1 \sim D+k$) and medium-term adequacy ($D+k+1 \sim D+N$), rather than managed by separate horizons.

9.7.2 Insight 2: Why Variable-Scale Scenario Metrics Are Needed

A single-scale uncertainty representation underestimates tail risk. The two-layer design explicitly separates: (i) frequent security fluctuations in Stage-2 and (ii) low-probability extreme adequacy events in Stage-3 including hydro drought. The C1 vs. C2 comparison shows that the small tree ($3 \times 2 = 6$ leaves) captures only modest load-shedding cost, while the full tree ($7 \times 5 = 35$ leaves) reveals 12.84 M¥—because it includes joint extreme events (load surge $\times 1.35$ + wind drought $\times 0.25$ + hydro drought $\times 0.35$). The graduated uncertainty model further sharpens risk assessment: day 2 sees only 40% of scenario deviation while day 3 absorbs the full deviation, creating asymmetric risk that drives RP and EV to different maintenance/ESS strategies. This improved risk visibility prevents under-preparation for extreme operating days.

9.7.3 Insight 3: Why Efficient Decoupling Is Operationally Necessary

With 35 leaf scenarios and binary recourse, the deterministic equivalent already reaches 70,001 variables and 24,369 binaries. As practical systems include more units, networks, and market constraints, decomposition methods become mandatory (§6). The convergence analysis shows that nested Benders guarantees LP-relaxation convergence finitely (Theorem 2), and exact MILP convergence is achievable via integer cuts (Proposition 4), while the gap bound (Theorem 3) provides a practical stopping criterion. Progressive hedging (Theorem 4) offers full parallelism across scenarios but remains a heuristic for the integer case (Remark 4).

9.7.4 Insight 4: Recommended Planning Workflow

For provincial operators, a pragmatic workflow is:

1. run deterministic EV for quick screening;
2. run stochastic RP with two-layer scenarios for final boundary decisions;
3. monitor VSS as a management KPI; if VSS exceeds a threshold (e.g., 3%–5%), enforce stochastic scheduling and activate decomposition mode.